FlowMLLab | Week 1.2

Pressure–Velocity Coupling

SIMPLE, PISO and PIMPLE

Finite-volume equations and the lid-driven cavity

Ehsan Roohi

The central problem. Momentum determines how fluid accelerates under pressure and viscous forces. Continuity constrains the resulting velocity. Pressure–velocity coupling obtains fields that satisfy both requirements on the same grid.

Prerequisites. Continuity and component momentum equations, finite differences, a linear system, and the Week 1 streamfunction–vorticity cavity example. Matrix notation is introduced after the scalar finite-volume equations.

Learning outcomes. Derive a cell pressure-correction equation; interpret predictor and corrected velocities; distinguish steady iteration, inner correction and outer correction; evaluate continuity, momentum defects, accuracy and computational work.

Sections	Topics
1–4	Conservation laws, finite volume, grid layout, momentum coefficients
5–8	Pressure correction, conservation example and SIMPLE
9–11	Backward Euler, PISO and a numerical correction example
12–15	PIMPLE, nonlinear coefficient updates and convergence
16–21	Recorded cavity comparisons and coupling study
22–23	Notebook controls, exercises and references

Scope. Two-dimensional, incompressible, laminar flow on a uniform staggered grid. The retained calculations compare $Re = 100, 400$ and 1000. Higher-Re spatial accuracy and transient accuracy require their own checks; convergence of a two-dimensional steady solver does not establish the physical stability of a three-dimensional flow.

Introduction A moving lid, a circulating fluid and an unknown pressure

Consider a square container filled with a viscous liquid. Its top wall moves to the right while the other three walls remain stationary. Friction with the moving wall starts the liquid moving. The walls prevent it from leaving the container, so the motion turns into a circulation inside the square. This is the *lid-driven cavity* problem.

To calculate that motion, we need both the fluid velocity and the pressure throughout the container. Two physical requirements determine them together.

Mass conservation. Imagine a small fixed region inside the liquid. For an incompressible liquid, whose density stays constant, the amount entering that region must equal the amount leaving it. Otherwise liquid would accumulate or disappear there. This requirement is called the *continuity equation*.

Momentum conservation. The liquid changes its motion because of forces. Pressure differences push it, and viscosity transfers motion between neighboring fluid layers. The *momentum equations* relate those effects to the fluid acceleration.

Neither requirement alone gives a complete solution. A velocity calculated with an inaccurate pressure can satisfy a momentum calculation while still giving unequal inflow and outflow in a small region. We therefore need a procedure that makes velocity and pressure satisfy both requirements.

The route through the lesson.

1. Write the continuity and momentum equations for this cavity.
2. Divide the domain into small cells and explain the finite-volume method: each cell must obey its own conservation balance.
3. Locate the stored pressure and velocity values and derive the equations that connect neighboring cells.
4. Explain how an initial pressure estimate is corrected. Then assemble these operations into SIMPLE, PISO and PIMPLE.
5. Examine calculated cavity fields, accuracy, convergence and measured cost.

The first calculation uses a Reynolds number of 100; its definition follows with the governing equations.

1 Incompressible flow: conservation laws

Horizontal velocity $u_d(x_d, y_d, t_d)$, vertical velocity $v_d(x_d, y_d, t_d)$ and pressure $p_d(x_d, y_d, t_d)$ are the unknown fields. Subscript d denotes dimensional quantities. For a Newtonian fluid with constant density ρ and constant kinematic viscosity ν, with no body force, continuity is

$$\frac{\partial u_d}{\partial x_d} + \frac{\partial v_d}{\partial y_d} = 0.$$

Conservation of horizontal and vertical momentum gives

$$\frac{\partial u_d}{\partial t_d} + u_d \frac{\partial u_d}{\partial x_d} + v_d \frac{\partial u_d}{\partial y_d} = -\frac{1}{\rho}\frac{\partial p_d}{\partial x_d} + \nu \left(\frac{\partial^2 u_d}{\partial x_d^2} + \frac{\partial^2 u_d}{\partial y_d^2} \right),$$
$$\frac{\partial v_d}{\partial t_d} + u_d \frac{\partial v_d}{\partial x_d} + v_d \frac{\partial v_d}{\partial y_d} = -\frac{1}{\rho}\frac{\partial p_d}{\partial y_d} + \nu \left(\frac{\partial^2 v_d}{\partial x_d^2} + \frac{\partial^2 v_d}{\partial y_d^2} \right).$$

The left sides contain local and convective acceleration. Pressure gradients and viscous diffusion provide the right-side accelerations. These two equations and continuity form the incompressible Navier–Stokes system.

Scales. Let L be cavity side length and U the lid speed. Define $x = x_d/L$, $y = y_d/L$, $t = t_d U/L$, $u = u_d/U$, $v = v_d/U$, and $p = (p_d - p_{\text{ref}})/(\rho U^2)$. Here p_{ref} is a fixed arbitrary reference pressure. Then $Re = UL/\nu$ and

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0,$$
$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right),$$
$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{Re}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right).$$

The following sections use these dimensionless quantities. At steady state the local time derivatives vanish, while convection, pressure and diffusion still balance.

Cavity conditions. The unit-square top lid has $u = 1, v = 0$; three stationary walls have $u = v = 0$. Every wall is impermeable. The ideal moving-lid corners contain a velocity discontinuity, so pressure and vorticity extrema there require care. The introductory case is $Re = 100$.

2 What the finite-volume method conserves

A *mesh* divides the fluid domain into small regions called cells. A *control volume* is a fixed region over which we impose a conservation balance. In this two-dimensional lesson it is a cell area with one unit of depth perpendicular to the page. A *face* is one boundary segment of that cell. Two neighboring cells have one face in common.

The finite-volume method starts from the conservation law for each control volume. It accounts for what is stored inside the region, what crosses its faces, and any source acting inside it. A *flux* is the amount transported across a surface per unit area and unit time. Multiplying by face area gives a transport rate. Here a face length times velocity gives volume flow rate per unit depth.

For the incompressible cavity, the mass inside a fixed cell is constant. Its continuity balance is therefore

$$\boxed{\text{total mass flow rate out of a cell} - \text{total mass flow rate into it} = 0.}$$

Momentum has a related balance, with acceleration, pressure forces and viscous transport as well as momentum carried by fluid crossing the faces.

How the equations become numerical. First integrate a conservation equation over the chosen region. Next express transport and forces through its faces. Finally approximate the required face values using a finite set of stored values. The resulting equation is an algebraic relation between neighboring unknowns. This process is called *discretization*.

Why sharing a face matters. Fluid leaving one cell through an internal face enters the neighboring cell through that same face. Using one numerical face flow rate in both balances makes these internal contributions cancel when the balances are added. Each individual cell must still balance; cancellation in the whole domain alone is insufficient.

2 continued Finite volume and finite differences

The finite-difference approach used in Week 1 begins with a differential equation evaluated at grid points. It replaces derivatives by combinations of nearby samples. For a smooth function $f(x)$ sampled at equally spaced positions x_i, define $f_i = f(x_i)$ and $h = x_{i+1} - x_i$. For example,

$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x_i} \simeq \frac{f_{i+1} - f_{i-1}}{2h}.$$

This is a centered approximation to a derivative *at a point*.

The finite-volume approach begins with a conservation balance *over a region*. Its approximations concern the values and gradients needed at the region's faces. Those gradients can also be evaluated with differences. Using a difference formula inside a finite-volume method does not change the fact that the starting equation is a cell conservation balance.

Question	Finite differences	Finite volume
Starting form	Differential equation at a grid location	Integrated conservation law for a cell
Numerical construction	Approximate derivatives from nearby samples	Approximate face transport and cell terms
Conservation check	Depends on the chosen discrete construction	Check the balance of every cell using shared face fluxes
This course	Week 1 streamfunction–vorticity solver	Week 1.2 pressure–velocity solver

On a uniform Cartesian mesh, the two approaches can produce identical algebraic formulas for some terms, such as constant-coefficient diffusion. Finite-difference schemes can also be constructed in conservative form. The distinction is the construction of the discrete equations, not a claim that one approach always conserves and the other never can.

For the cavity, the next pages choose where to store each unknown. Then the continuity integral is evaluated over one cell and the momentum integral over a region associated with one velocity unknown. Their locations and boundaries are specified before performing those integrations.

2 continued Staggered locations and a pressure control volume

The equations in Section 1 are dimensionless. We now integrate them over a small fixed region of the grid. A *control volume* is a chosen region of the flow domain. In a two-dimensional calculation it is an area per unit depth in the third direction. Pressure p_P is stored at pressure-cell center P. Horizontal velocity is stored at vertical faces; vertical velocity is stored at horizontal faces. For the cell around P, use lowercase e, w, n, s for east, west, north and south faces. Uppercase E, W, N, S denote neighboring *pressure centers*. Here a pressure center means the cell center at which pressure is stored. The *shared velocity face* between P and E is their common vertical boundary; the horizontal velocity stored there is u_e. It is one value used in the continuity balances of both cells. This storage pattern is the MAC (marker-and-cell) staggered grid: pressure and the two velocity components occupy different locations.

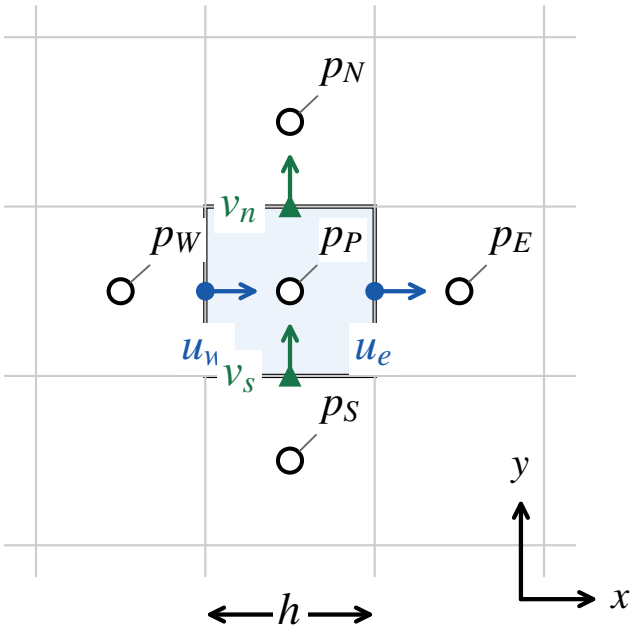


Figure 1: Staggered (MAC) storage. Open black circles mark cell centers where pressure is stored; blue filled circles mark horizontal velocities and green triangles mark vertical velocities. The shaded square is the pressure control volume around P, of side length h.

2 continued Geometry and outward directions of one cell

First allow a rectangular cell:

$$V_P = [x_w, x_e] \times [y_s, y_n], \qquad \Delta x = x_e - x_w, \quad \Delta y = y_n - y_s, \quad |V_P| = \Delta x \Delta y.$$

Here $|V_P|$ means its area. The boundary is ∂V_P. Its outward unit normal $\boldsymbol{n}$ points away from P at each face:

Face	Outward normal	Face length	Outward velocity component
East	$(1, 0)$	Δy	$+u_e$
West	$(-1, 0)$	Δy	$-u_w$
North	$(0, 1)$	Δx	$+v_n$
South	$(0, -1)$	Δx	$-v_s$

All u values are positive rightward and all v values positive upward. These coordinate conventions do not reverse at the west or south face. The minus signs arise from the *outward normal*, rather than from a different definition of velocity.

The uniform square grid used below has $\Delta x = \Delta y = h = 1/N$, where N is the number of pressure cells along one side. Its cell area is h^2 and each face length per unit depth is h.

2 continued Why pressure and velocity are staggered

In the preceding grid, a cell center is where pressure is stored. The shared face is the boundary between two adjacent cells, where the horizontal velocity is stored. Along one horizontal row, let x_i be center i's position and p_i its pressure sample. The midpoint $x_{i+1/2} = (x_i + x_{i+1})/2$ is the position of the face between centers i and $i + 1$.

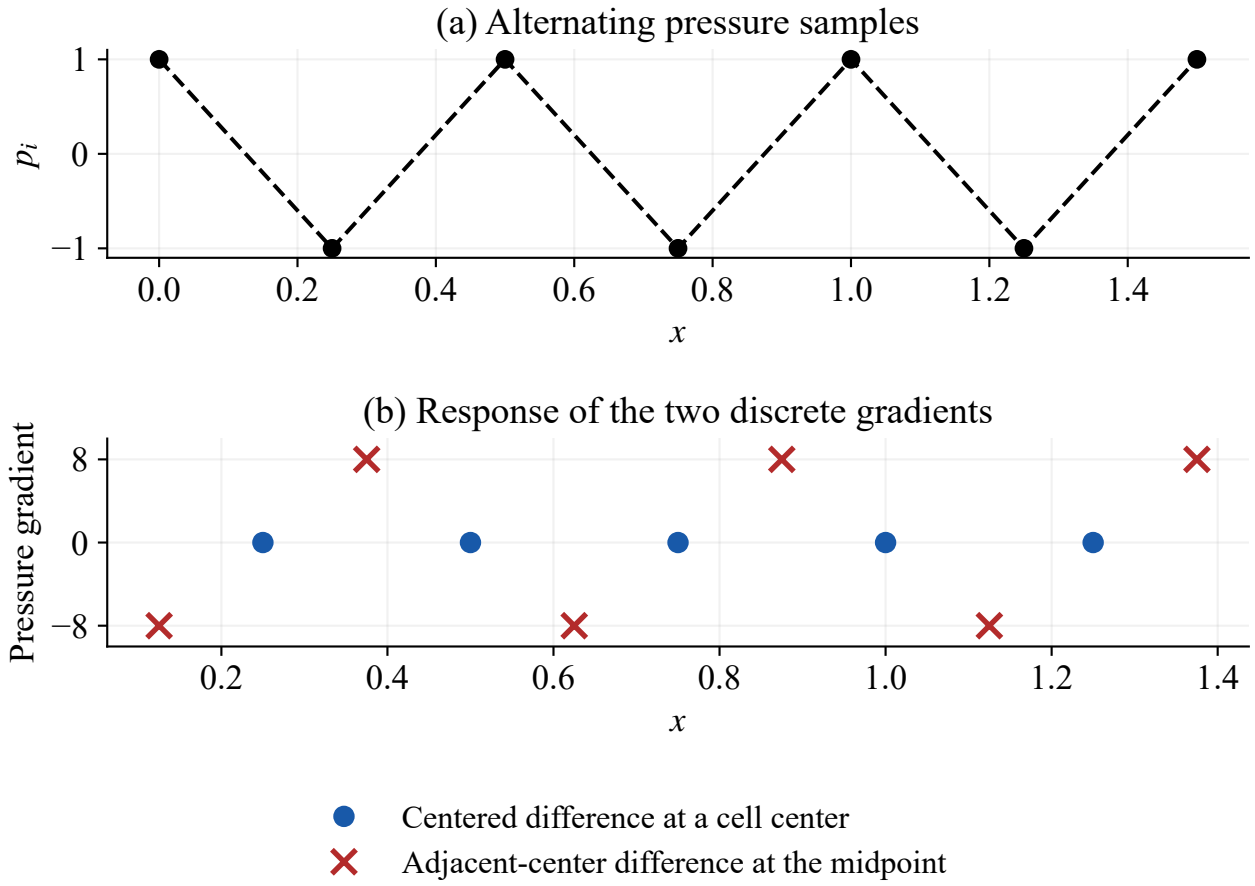


Figure 2: A synthetic pressure diagnostic with h=0.25. (a) Pressure samples at equally spaced cell centers alternate between +1 and -1. (b) Blue circles use the two samples on either side of a cell center; red crosses use the two adjacent samples around their midpoint. The blue result is zero while the red results are +8 or -8. The legend identifies these two difference formulas and their evaluation positions; it does not show two flow solutions.

Panel (b) compares *where* a pressure derivative is evaluated and *which pair* of stored values is used. Blue circles refer to a centered derivative at a cell center; red crosses refer to a derivative at the midpoint between adjacent centers. Their definitions are expanded next.

2 continued Read the two pressure-gradient formulas

Consider a synthetic interior grid patch with $p_i = (-1)^i$. The index i counts pressure samples along x; the spacing is $h = 0.25$. Neighboring pressures alternate between $+1$ and -1.

If velocity and pressure were both stored at the same centers and the pressure gradient were approximated by a naive central difference there, it would give

$$\left(\frac{\partial p}{\partial x}\right)_i \simeq \frac{p_{i+1} - p_{i-1}}{2h} = 0,$$

because $p_{i+1} = p_{i-1}$. This alternating pressure mode is invisible to that gradient: the resulting momentum equation cannot distinguish it from zero pressure gradient in the interior.

On the MAC grid, horizontal momentum is evaluated at the shared face between adjacent pressure centers. Its pressure gradient is

$$\left(\frac{\partial p}{\partial x}\right)_{i+1/2} \simeq \frac{p_{i+1} - p_i}{h} = \pm\frac{2}{h} = \pm 8.$$

The face momentum equation responds to the adjacent pressure difference. This is one reason for staggering the unknowns. In two dimensions an alternating pattern in both directions is called a pressure checkerboard.

The figure is a discrete diagnostic pattern, not a computed cavity solution. Properly stabilized collocated methods can also work; they require their own face-flux interpolation treatment. The present course implementation uses the staggered arrangement derived here.

2 continued Integral and discrete continuity equations

Start with the differential equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

Integrate both terms over the same cell, with $\mathrm{d}V = \mathrm{d}x\,\mathrm{d}y$:

$$\int_{y_s}^{y_n} \int_{x_w}^{x_e} \frac{\partial u}{\partial x}\,\mathrm{d}x\,\mathrm{d}y + \int_{x_w}^{x_e} \int_{y_s}^{y_n} \frac{\partial v}{\partial y}\,\mathrm{d}y\,\mathrm{d}x = 0.$$

For the first term integrate in x while holding y fixed. For the second integrate in y while holding x fixed. The fundamental theorem of calculus gives

$$\int_{y_s}^{y_n} [u(x_e, y) - u(x_w, y)]\,\mathrm{d}y + \int_{x_w}^{x_e} [v(x, y_n) - v(x, y_s)]\,\mathrm{d}x = 0.$$

Thus a derivative inside the cell becomes the difference of values on its two opposite faces. Write the velocity vector as $\boldsymbol{u} = (u, v)$ and the outward normal as $\boldsymbol{n} = (n_x, n_y)$. Their dot product $\boldsymbol{u}\cdot\boldsymbol{n} = un_x + vn_y$ is the outward normal speed. The divergence is $\nabla \cdot \boldsymbol{u} = \partial u/\partial x + \partial v/\partial y$. With $\mathrm{d}S$ denoting a boundary-length element, the same identity is the divergence theorem in two dimensions:

$$\int_{V_P} \nabla \cdot \boldsymbol{u}\,\mathrm{d}V = \int_{\partial V_P} \boldsymbol{u} \cdot \boldsymbol{n}\,\mathrm{d}S = 0.$$

Define each face-average velocity by its face integral; for example,

$$\overline{u}_e = \frac{1}{\Delta y} \int_{y_s}^{y_n} u(x_e, y)\,\mathrm{d}y.$$

Replacing a face average by the stored face-center sample is the midpoint approximation used on this grid. With those samples the discrete balance is

$$\underbrace{\Delta y\, u_e}_{\text{east outflow}} - \underbrace{\Delta y\, u_w}_{\text{west inflow}} + \underbrace{\Delta x\, v_n}_{\text{north outflow}} - \underbrace{\Delta x\, v_s}_{\text{south inflow}} = 0.$$

Each term is a signed volume flux per unit depth. Dividing by cell area gives

$$\frac{u_e - u_w}{\Delta x} + \frac{v_n - v_s}{\Delta y} = 0, \qquad \boxed{r_{c,P} = \frac{u_e - u_w + v_n - v_s}{h} = 0 \quad \text{on a square grid.}}$$

The residual $r_{c,P}$ is net outward volume flux divided by area. It is not a pressure change. For constant density ρ, multiplication of every flux by ρ gives the corresponding mass balance.

Numerical check. With $h = 0.25$ and $(u_e, u_w, v_n, v_s) = (0.30, 0.20, 0.10, 0.05)$, net outward flux is $0.25(0.30 - 0.20 + 0.10 - 0.05) = 0.0375$ and $r_{c,P} = 0.0375/0.25^2 = 0.60$. This tentative velocity violates local continuity even if a neighboring cell has an equal and opposite imbalance. Shared internal fluxes cancel in a global sum; they must still balance in each individual cell.

3 Conservative form and integral momentum balance

The horizontal momentum equation from Section 1 is

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re}\nabla^2 u.$$

To convert convection into face fluxes, apply the product rule:

$$\begin{aligned}\frac{\partial(u^2)}{\partial x} + \frac{\partial(uv)}{\partial y} &= 2u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + u\frac{\partial v}{\partial y} \\ &= u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + u\underbrace{\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)}_{0 \text{ by continuity}}.\end{aligned}$$

The gradient $\nabla u = (\partial u/\partial x, \partial u/\partial y)$ collects the two spatial derivatives of the horizontal velocity. Its divergence is $\nabla\cdot(\nabla u) = \partial^2 u/\partial x^2 + \partial^2 u/\partial y^2$, the Laplacian in the viscous term. Consequently, the incompressible equation is

$$\frac{\partial u}{\partial t} + \nabla\cdot(u\boldsymbol{u}) = -\frac{\partial p}{\partial x} + \frac{1}{Re}\nabla\cdot(\nabla u).$$

The scalar u in $u\boldsymbol{u}$ is the horizontal momentum per unit mass being transported; $\boldsymbol{u} = (u, v)$ is the transporting velocity.

Integrate over the *horizontal-velocity* control volume V_e, centered on the stored velocity u_e. It is a different region from pressure cell V_P. For a fixed control volume, the time derivative can pass outside the integral. Apply the divergence theorem separately to convection, pressure and diffusion:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{V_e} u\,\mathrm{d}V + \int_{\partial V_e} u(\boldsymbol{u}\cdot\boldsymbol{n})\,\mathrm{d}S = -\int_{\partial V_e} p\,n_x\,\mathrm{d}S + \frac{1}{Re}\int_{\partial V_e}\nabla u\cdot\boldsymbol{n}\,\mathrm{d}S.$$

Here n_x is the horizontal component of the outward unit normal. The pressure identity used above is

$$\int_{V_e}\frac{\partial p}{\partial x}\,\mathrm{d}V = \int_{\partial V_e} p\,n_x\,\mathrm{d}S.$$

Term	Physical meaning in this equation
Time derivative	Accumulation of horizontal momentum per unit density
Convective integral	Horizontal momentum carried through the boundary by flow
Pressure integral	Horizontal pressure force per unit density
Diffusive integral	Viscous transport of horizontal momentum

These integral balances follow from the differential equations. Face quadrature and interpolation convert them into equations for the stored grid values.

3 continued The momentum volume and its four face contributions

The horizontal velocity u_e lies halfway between pressure centers P and E. Its square momentum volume extends from x_P to x_E, a distance h. The pressure centers therefore lie at the midpoints of its west and east faces.

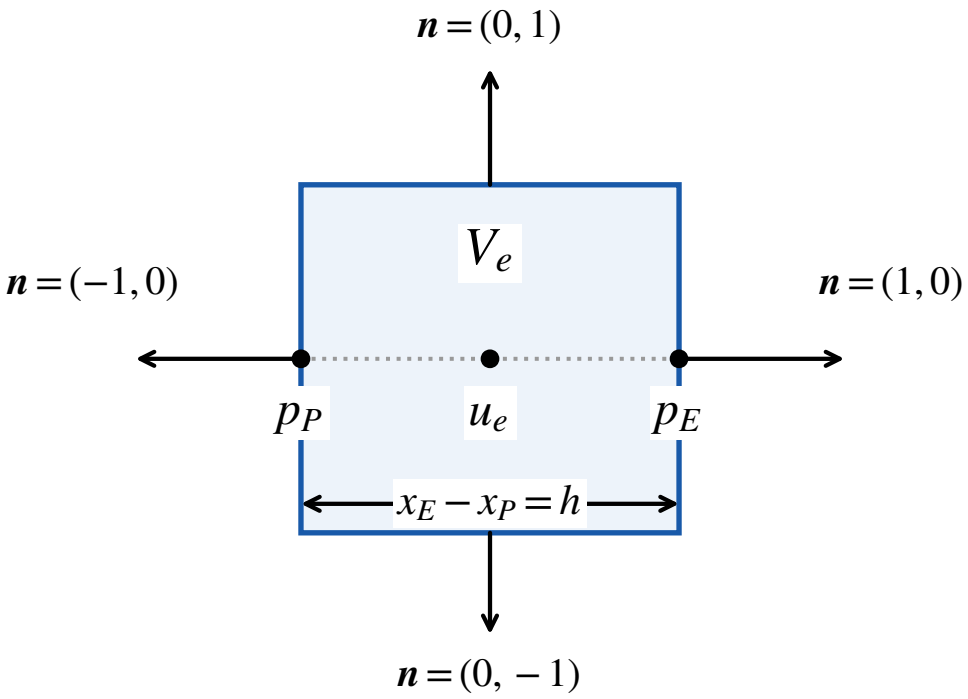


Figure 3: Horizontal-momentum control volume V_e. The unknown u_e is at its center, while p_P and p_E are at the west and east face midpoints. Arrows show outward unit normals. The two pressure samples are separated by h.

At this volume's boundaries, write $c_\mathrm{E}, c_\mathrm{W}$ for the horizontal advecting velocities and $c_\mathrm{N}, c_\mathrm{S}$ for the vertical advecting velocities. All are positive in their coordinate directions. Write $\widehat{u}_\mathrm{E}$, $\widehat{u}_\mathrm{W}$, $\widehat{u}_\mathrm{N}$, $\widehat{u}_\mathrm{S}$ for the horizontal component being *transported* at those boundaries. The hat values are interpolated from neighboring velocity samples.

Convection. Midpoint evaluation of the four integrals gives

$$h\left(c_\mathrm{E}\widehat{u}_\mathrm{E} - c_\mathrm{W}\widehat{u}_\mathrm{W} + c_\mathrm{N}\widehat{u}_\mathrm{N} - c_\mathrm{S}\widehat{u}_\mathrm{S}\right).$$

North/south transport is still transport of *horizontal* momentum: the fluid crosses those faces with vertical velocity while carrying horizontal u.

Pressure. Only the east and west faces have $n_x \neq 0$. Approximating their pressure averages by p_E and p_P gives

$$-\int_{\partial V_e} p n_x \,\mathrm{d}S \simeq -h p_E + h p_P = -h(p_E - p_P).$$

After dividing by area h^2, this is $-(p_E - p_P)/h$. The separation between P and E is h, not $2h$.

Diffusion. The normal derivative changes sign on west and south:

$$\frac{h}{Re}\left[\left(\frac{\partial u}{\partial x}\right)_\mathrm{E} - \left(\frac{\partial u}{\partial x}\right)_\mathrm{W} + \left(\frac{\partial u}{\partial y}\right)_\mathrm{N} - \left(\frac{\partial u}{\partial y}\right)_\mathrm{S}\right].$$

Accumulation. Approximate the volume integral by $h^2 u_e$, so its derivative is $h^2\,\mathrm{d}u_e/\mathrm{d}t$. It vanishes for the target steady problem. Section 9 introduces the discrete time derivative for a transient calculation.

3 continued Locate the velocity before interpolating it

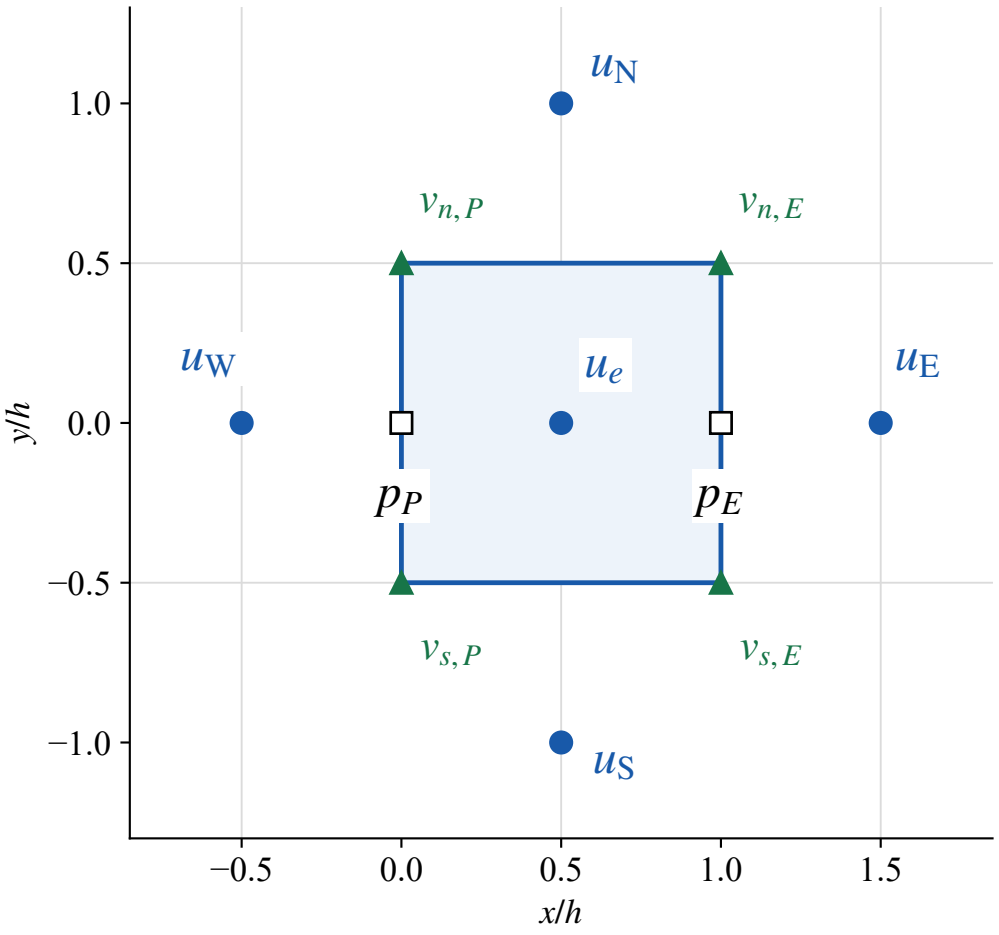


Figure 4: Stored samples around the horizontal-momentum volume (blue square). Blue circles: horizontal velocity; green triangles: vertical velocity; open black squares: pressure. Coordinates are relative to pressure center P and scaled by h.

Use local coordinates with origin at pressure center P. The axes are measured in grid spacings h. The pressure centers are $P = (0, 0)$ and $E = (h, 0)$. The horizontal velocity u_e is stored at $(h/2, 0)$, on their shared pressure-cell face. Its own momentum control volume is shifted: $0 \leq x \leq h$ and $-h/2 \leq y \leq h/2$.

The blue circles are stored horizontal components. For example, u_N is a neighboring horizontal component one spacing above u_e, not a vertical velocity. Green triangles mark stored vertical components. Their averages supply the speeds carrying fluid through the north and south momentum faces.

3 continued Interpolate the transported horizontal component

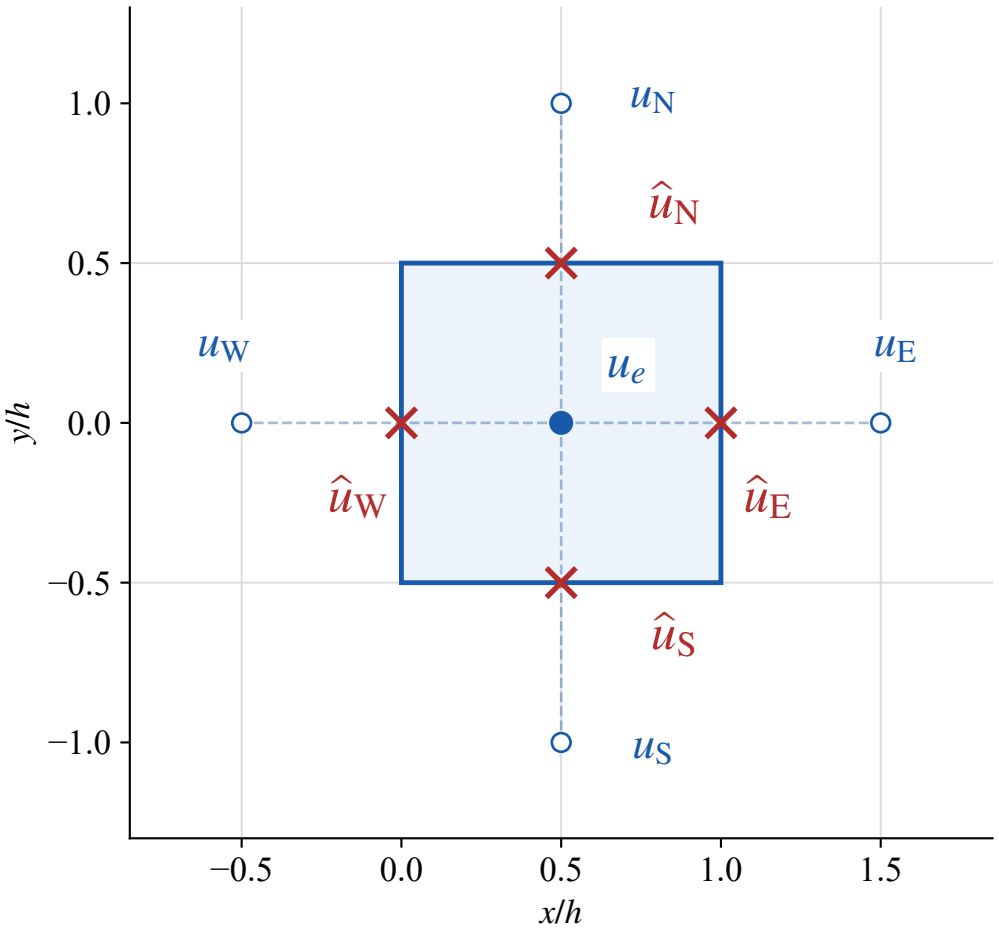


Figure 5: Evaluation points for transported horizontal velocity. Blue circles are stored u samples and red crosses are interpolated values on momentum-volume faces. Dashed segments connect each pair used in an average. The west and east crosses coincide geometrically with pressure centers; pressure symbols are omitted here for clarity.

The red crosses are evaluation points at the faces of the horizontal-momentum volume. In these face-flux formulas, a hat denotes interpolation to a momentum-volume face. The east and west crosses coincide with pressure centers; the north and south crosses lie at pressure-cell corners. These geometrical coincidences do not make pressure and horizontal velocity the same variable.

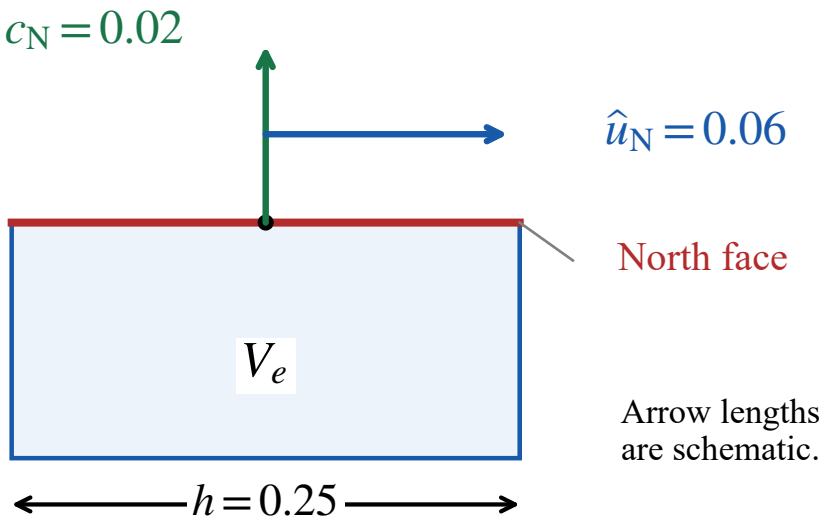
Each cross lies halfway between two blue samples. For example,

$$\widehat{u}_N = \frac{u_e + u_N}{2}, \qquad \widehat{u}_E = \frac{u_e + u_E}{2}.$$

For example, the north cross is at $(h/2, h/2)$, halfway between u_e at $(h/2, 0)$ and u_N at $(h/2, h)$. The same construction applies to the east, west and south faces.

Read each mark in its equation. In the face-flux formulas a hat indicates interpolation; a star indicates a provisional momentum solution; a prime indicates a pressure or velocity correction. None of these marks independently identifies a pressure-cell center. The later PIMPLE equations use $q_{\text{lag}}^{(\ell)}$ and $p_{\text{lag}}^{(\ell)}$ for the latest outer-iteration estimates, keeping the hat for spatial interpolation.

3 continued Crossing a face while carrying horizontal momentum



$$Q_N = hc_N = 0.25(0.02) = 0.005$$

$$F_N^u = Q_N \hat{u}_N = 0.005(0.06) = 0.00030$$

Figure 6: Horizontal momentum transported through the north face. The normal speed c_N carries fluid upward through the red face; the transported component is the horizontal velocity $\hat{u}_N$, offset above the face for clarity. Q_N is outward volume flux per unit depth. F_N^u is horizontal-momentum flux per unit density and unit depth. The values are nondimensional; arrow lengths are schematic.

At the north face, c_N is vertical: it carries fluid across the face. The transported quantity $\hat{u}_N$ is horizontal. Per unit density and unit depth, their product times the face length is the integrated horizontal-momentum flux $hc_N\hat{u}_N$.

Arithmetic example. Take nondimensional $h = 0.25$ and the stored values

$$(u_e, u_E, u_W, u_N, u_S) = (0.04, 0.12, 0.02, 0.08, -0.02).$$

The four averages are

$$(\hat{u}_E, \hat{u}_W, \hat{u}_N, \hat{u}_S) = (0.08, 0.03, 0.06, 0.01).$$

Let the normal advecting speeds be $(c_E, c_W, c_N, c_S) = (0.08, 0.03, 0.02, 0.07)$. The north value averages $v_{n,P} = 0.03$ and $v_{n,E} = 0.01$; the south value averages $v_{s,P} = 0.05$ and $v_{s,E} = 0.09$. These values also satisfy continuity in the adjacent pressure cells P and E. The momentum volume's outward volume-flux balance is $h(0.08 - 0.03 + 0.02 - 0.07) = 0$. Its horizontal-momentum balance is different:

$$\begin{aligned} h(c_E\hat{u}_E - c_W\hat{u}_W + c_N\hat{u}_N - c_S\hat{u}_S) &= 0.001600 - 0.000225 + 0.000300 - 0.000175 \\ &= 0.001500. \end{aligned}$$

Dividing by area h^2 gives the convection term 0.024. The notebook checks these values. This is a local interpolation example, not a solved cavity field.

3 continued Central interpolation and face gradients

In one linear momentum solve, advecting speeds c are evaluated from the known velocity estimate and then held fixed. This is called freezing the coefficients. The transported horizontal components remain unknowns. At east and west their averaging formulas have the same geometry, but their roles in the solve differ.

Let $u_{\mathrm{E}}, u_{\mathrm{W}}, u_{\mathrm{N}}, u_{\mathrm{S}}$ be neighboring stored *horizontal* velocities one grid spacing from u_e. They are not the pressure samples p_E, p_W, p_N, p_S. For central interpolation,

$$\widehat{u}_{\mathrm{E}} = (u_e + u_{\mathrm{E}})/2, \qquad \widehat{u}_{\mathrm{W}} = (u_{\mathrm{W}} + u_e)/2,$$
$$\widehat{u}_{\mathrm{N}} = (u_e + u_{\mathrm{N}})/2, \qquad \widehat{u}_{\mathrm{S}} = (u_{\mathrm{S}} + u_e)/2.$$

The advecting velocities are obtained from the staggered grid:

$$c_{\mathrm{E}} = (u_e + u_{\mathrm{E}})/2, \qquad c_{\mathrm{W}} = (u_{\mathrm{W}} + u_e)/2,$$
$$c_{\mathrm{N}} = (v_{n,P} + v_{n,E})/2, \qquad c_{\mathrm{S}} = (v_{s,P} + v_{s,E})/2.$$

For example, $v_{n,P}$ and $v_{n,E}$ are the vertical velocities on the north faces of pressure cells P and E. During a linear momentum solve the c values are held at their current estimates; updating them is a later nonlinear step.

Approximate the gradients at the momentum-volume faces by

$$u_{x,\mathrm{E}} \simeq (u_{\mathrm{E}} - u_e)/h, \qquad u_{x,\mathrm{W}} \simeq (u_e - u_{\mathrm{W}})/h,$$
$$u_{y,\mathrm{N}} \simeq (u_{\mathrm{N}} - u_e)/h, \qquad u_{y,\mathrm{S}} \simeq (u_e - u_{\mathrm{S}})/h.$$

Here $u_{x,\mathrm{E}}$ means $\partial u/\partial x$ evaluated at the east face. Substitution and division by h^2 turn diffusion into

$$\beta(u_{\mathrm{E}} + u_{\mathrm{W}} + u_{\mathrm{N}} + u_{\mathrm{S}} - 4u_e), \qquad \boxed{\beta = \frac{1}{Re\, h^2}.}$$

For instance, the east and west contributions combine as $[(u_{\mathrm{E}} - u_e) - (u_e - u_{\mathrm{W}})]/(Re\, h^2)$. The two occurrences of $-u_e$ explain the central $-2u_e$ in that direction.

The full volume-normalized equation is therefore

$$\begin{aligned}\frac{\mathrm{d}u_e}{\mathrm{d}t} &+ \frac{c_{\mathrm{E}}(u_e + u_{\mathrm{E}}) - c_{\mathrm{W}}(u_{\mathrm{W}} + u_e)}{2h} \\ &+ \frac{c_{\mathrm{N}}(u_e + u_{\mathrm{N}}) - c_{\mathrm{S}}(u_{\mathrm{S}} + u_e)}{2h} \\ &= -\frac{p_E - p_P}{h} + \beta(u_{\mathrm{E}} + u_{\mathrm{W}} + u_{\mathrm{N}} + u_{\mathrm{S}} - 4u_e).\end{aligned}$$

Each term has the form of an acceleration. Dividing face length h by cell area h^2 accounts for the factors of $1/h$ in these terms.

3 continued Momentum coefficients and numerical example

For the steady interior row, move each convective neighbor term to the right and each central diffusive term to the left. Introduce a_e for the coefficient of the velocity being solved, $a_\mathrm{E}, a_\mathrm{W}, a_\mathrm{N}, a_\mathrm{S}$ for its four neighbor coefficients, and b_e for known terms collected on the right side, such as a prescribed wall-velocity contribution. The result is

$$\boxed{a_e u_e = a_\mathrm{E} u_\mathrm{E} + a_\mathrm{W} u_\mathrm{W} + a_\mathrm{N} u_\mathrm{N} + a_\mathrm{S} u_\mathrm{S} + b_e - \frac{p_E - p_P}{h}.}$$

An interior steady row has no time or wall source, so $b_e = 0$ there. The neighbor coefficients are

$$a_\mathrm{E} = \beta - c_\mathrm{E}/(2h), \qquad a_\mathrm{W} = \beta + c_\mathrm{W}/(2h),$$
$$a_\mathrm{N} = \beta - c_\mathrm{N}/(2h), \qquad a_\mathrm{S} = \beta + c_\mathrm{S}/(2h).$$

For example, the east contribution to convection is $c_\mathrm{E} u_e/(2h) + c_\mathrm{E} u_\mathrm{E}/(2h)$. The east diffusion term on the right is $\beta u_\mathrm{E} - \beta u_e$. Moving terms gives $(\beta - c_\mathrm{E}/(2h))u_\mathrm{E}$ on the right and $(\beta + c_\mathrm{E}/(2h))u_e$ on the left. Repeating the operation at all faces gives

$$\begin{aligned} a_e &= 4\beta + \frac{c_\mathrm{E} - c_\mathrm{W} + c_\mathrm{N} - c_\mathrm{S}}{2h} \\ &= a_\mathrm{E} + a_\mathrm{W} + a_\mathrm{N} + a_\mathrm{S} + \frac{c_\mathrm{E} - c_\mathrm{W} + c_\mathrm{N} - c_\mathrm{S}}{h}. \end{aligned}$$

Do not set the last term to zero merely because the target flow is incompressible: the current advecting estimates can still have a discrete mass imbalance. Pressure–velocity coupling removes that imbalance iteratively. Known wall values contribute to the source and can modify the diagonal. For a transient problem, the time derivative also supplies a center term and a known old-time source; those are derived in Section 9.

One frozen-row example. Set $h = 0.25$, $Re = 100$, $c_\mathrm{E} = c_\mathrm{W} = 0.02$, $c_\mathrm{N} = c_\mathrm{S} = 0$. Then $\beta = 0.16$ and

$$(a_\mathrm{E}, a_\mathrm{W}, a_\mathrm{N}, a_\mathrm{S}) = (0.12, 0.20, 0.16, 0.16), \qquad a_e = 0.64.$$

For given neighbor values $(u_\mathrm{E}, u_\mathrm{W}, u_\mathrm{N}, u_\mathrm{S}) = (0.04, 0.02, 0.03, 0.01)$, their weighted sum is

$$0.12(0.04) + 0.20(0.02) + 0.16(0.03) + 0.16(0.01) = 0.0152.$$

With $p_P = 0.010$, $p_E = 0.008$ and $b_e = 0$, the pressure contribution is $-(0.008 - 0.010)/0.25 = +0.008$. Thus

$$\boxed{u_e = (0.0152 + 0.008)/0.64 = 0.03625.}$$

The lower east pressure drives a positive horizontal contribution. This satisfies one frozen momentum row; it does not by itself enforce the neighboring pressure cells' continuity equations. That is the purpose of SIMPLE. For vertical momentum, apply the same construction around v_n; its pressure term is $-(p_N - p_P)/h$. The next section assembles these scalar rows into a matrix.

4 Read one discrete momentum equation

The previous section produced an equation for the horizontal velocity u_e. It has one unknown at the center of a horizontal-momentum control volume and four neighboring horizontal velocities. Before shortening that equation, identify its entries:

Entry	Meaning in this row
u_e	Horizontal velocity being calculated. It is stored at face e between pressure cells P and E.
u_E, u_W, u_N, u_S	Horizontal velocities one mesh spacing east, west, north and south of u_e.
a_e	Center coefficient multiplying u_e. It comes from discretized convection and diffusion.
a_E, a_W, a_N, a_S	Coefficients multiplying the four neighboring velocities.
b_e	Known source contribution after prescribed boundary values have been collected on the right side.
p_P, p_E	Pressures at adjacent cell centers, separated by h.

The row is

$$a_e u_e = a_E u_E + a_W u_W + a_N u_N + a_S u_S + b_e - \frac{p_E - p_P}{h}.$$

The neighbor terms describe interaction with the surrounding flow. The last term is the horizontal pressure force per unit mass. All rows here have already been divided by control-volume area; their terms have the dimensionless form of accelerations.

Where the source comes from. At a steady interior location, there is no prescribed wall value in the row and this cavity has no body-force source: $b_e = 0$. At the moving lid, the stored velocity is a distance $h/2$ below the prescribed wall velocity U_{wall}. Approximate the normal diffusive gradient using that half-cell distance. The north-face contribution, divided by the momentum-volume area, is

$$\frac{1}{Re}\frac{h}{h^2}\frac{U_{\text{wall}} - u_e}{h/2} = 2\beta U_{\text{wall}} - 2\beta u_e, \qquad \beta = \frac{1}{Re\,h^2}.$$

Move $-2\beta u_e$ to the left: it contributes $+2\beta$ to the diagonal. The known term $2\beta U_{\text{wall}}$ stays on the right and enters b_e. For $Re = 100$, $h = 0.25$ and $U_{\text{wall}} = 1$, these wall contributions are 0.32 to the diagonal and 0.32 to the source. The complete wall row also includes its remaining face contributions. The symbol b_e collects known terms; it is not another unknown. Time-dependent sources are introduced with the time discretization in Section 9.

4 continued A source, a neighbor sum and one velocity

The quantity needed from the neighboring velocities is their weighted sum plus the known source. Call it H_e:

$$H_e = b_e + a_{\mathrm{E}}u_{\mathrm{E}} + a_{\mathrm{W}}u_{\mathrm{W}} + a_{\mathrm{N}}u_{\mathrm{N}} + a_{\mathrm{S}}u_{\mathrm{S}}.$$

Define $d_e = 1/a_e$, the reciprocal of the center coefficient. Dividing the momentum row by a_e now gives

$$\boxed{u_e = d_e\left(H_e - \frac{p_E - p_P}{h}\right).}$$

For fixed coefficients, changing a neighbor velocity changes H_e. Changing the pressure difference changes the pressure contribution. These are the two pieces used repeatedly in pressure–velocity coupling.

Use the preceding interior example: $a_e = 0.64$, $b_e = 0$, and $(a_{\mathrm{E}}, a_{\mathrm{W}}, a_{\mathrm{N}}, a_{\mathrm{S}}) = (0.12, 0.20, 0.16, 0.16)$. With neighboring velocities $(0.04, 0.02, 0.03, 0.01)$,

$$H_e = 0 + 0.12(0.04) + 0.20(0.02) + 0.16(0.03) + 0.16(0.01) = 0.0152.$$

For $p_P = 0.010$, $p_E = 0.008$ and $h = 0.25$,

$$\frac{p_E - p_P}{h} = -0.008, \qquad d_e = 1/0.64 = 1.5625,$$
$$u_e = 1.5625[0.0152 - (-0.008)] = 0.03625.$$

This is the same scalar momentum calculation, with its terms grouped under two names. It does not yet enforce the four-face continuity balance in each pressure cell.

Generalizing the location. Let f label any unknown velocity location and write its component as q_f: either a horizontal u or a vertical v. Let g label a neighboring location contributing to that same component row. The set of such neighbors is $\mathrm{NB}(f)$, and a_{fg} multiplies q_g. Then

$$H_f = b_f + \sum_{g \in \mathrm{NB}(f)} a_{fg} q_g, \qquad d_f = 1/a_f.$$

The sum adds the neighbor contributions just evaluated explicitly. The generic symbol q_f represents a velocity component, not a volume flow rate.

4 continued Collect the scalar rows into a linear system

A *vector* is an ordered column of values. Collect all unknown interior horizontal and vertical face velocities into the column q. Collect the cell-center pressures into p and the known source entries b_f into b. Use the same ordering every time these columns are assembled.

A *matrix* stores the coefficients multiplying that column of unknowns. For the momentum matrix A, put a_f on row f's diagonal. Move the neighbor terms to the left, so its neighbor entries are $-a_{fg}$:

$$(Aq)_f = a_f q_f - \sum_{g \in \mathrm{NB}(f)} a_{fg} q_g.$$

The notation $(Aq)_f$ means entry f of the product Aq. It is precisely the left side of one rearranged scalar momentum equation.

Define G as the discrete pressure-gradient operation. It maps the stored pressures to a pressure gradient at each unknown velocity location. For example,

$$(Gp)_e = \frac{p_E - p_P}{h}, \qquad (Gp)_n = \frac{p_N - p_P}{h}.$$

The first entry acts in horizontal momentum and the second in vertical momentum. The complete set of rows is therefore

$$\boxed{Aq = b - Gp.}$$

This abbreviates the equations already derived. Once coefficients, boundary values and a pressure estimate are specified, a linear solver calculates the velocity column.

For the square mesh with N pressure cells along each side, there are N^2 pressures. Unknown interior velocities comprise $N(N-1)$ horizontal and $N(N-1)$ vertical values. Thus q and b each have $2N(N-1)$ entries, and A has that many rows and columns. Prescribed wall-normal velocities are boundary data, rather than entries in q. The Python arrays retain those boundary values as well: pressure has shape (N, N), horizontal velocity $(N, N+1)$ and vertical velocity $(N+1, N)$. Array entry $[j, i]$ uses row j for the y direction and column i for x. The column q used in the algebra contains only the interior unknown entries extracted from those full arrays.

4 continued The operators used by the coupling algorithms

Define the discrete divergence operation D by applying the derived cell continuity balance to the stored face velocities:

$$(Dq)_P = \frac{u_e - u_w + v_n - v_s}{h} = r_{c,P}.$$

Thus Dq is a column of cell imbalances, one entry per pressure cell. Zero in every entry means discrete continuity. The gradient G maps pressure values to face gradients; D maps face velocities back to cell balances.

Call the diagonal part of the momentum matrix M, the remaining part R, and the inverse of the diagonal d:

$$M = \mathrm{diag}(a_f), \qquad R = A - M, \qquad d = M^{-1} = \mathrm{diag}(1/a_f).$$

The notation $\mathrm{diag}(a_f)$ means a matrix containing the center coefficients on its diagonal and zeros elsewhere. The inverse d simply divides entry f by a_f. The matrix R contains the off-diagonal entries $-a_{fg}$, so

$$H(q) = b - Rq, \qquad q = d[H(q) - Gp].$$

In entry f, $-Rq$ restores the positive neighbor sum used in H_f. The notation $H(q)$ emphasizes that these contributions change when the neighbor velocities change. Coefficients stay fixed during one linear solve; convection coefficients are rebuilt between nonlinear updates.

Symbol	Operation and input/output
A, b	Momentum coefficients and known sources; $Aq = b - Gp$.
G	Cell pressures in; pressure gradients at velocity faces out.
D	Face velocities in; cell continuity residuals out.
d	Scale each velocity-row entry by its reciprocal center coefficient.
R	Off-diagonal momentum coefficients, carrying neighbor coupling.
$H(q)$	Known sources plus neighbor-velocity contributions.

Pressure is solved at cell centers and velocity at faces, so D and G map between different collections of values. Section 6 derives the pressure matrix from these operations. The small letter d is distinct from the grid spacing h.

5 Why a momentum solution needs a pressure correction

Suppose the velocity and pressure estimates have already been updated k times. Superscript k labels that steady coupling iteration. It is not a physical time level. The current pressure column is p^k.

Hold the momentum coefficients fixed and solve

$$Aq^* = b - Gp^k.$$

The star labels the *predictor*: the velocity obtained using this pressure estimate. It is provisional because the momentum solve alone does not impose continuity in every cell. Compute its cell residuals from Dq^*.

For one square pressure cell with $h = 0.25$, suppose its predicted velocities are

$$(u_e^*, u_w^*, v_n^*, v_s^*) = (0.12, 0.08, 0.01, 0.03).$$

The star on each component has the same predictor meaning. The integrated outward volume flow rate per unit depth, called Q_P^*, is

$$Q_P^* = h(u_e^* - u_w^* + v_n^* - v_s^*) = 0.25(0.12 - 0.08 + 0.01 - 0.03) = 0.005.$$

Dividing by cell area gives the continuity residual,

$$r_{c,P}^* = Q_P^*/h^2 = 0.005/0.0625 = 0.08.$$

More fluid leaves than enters. The face velocities must change so that their combined outward flow changes by –0.005. Pressure correction is obtained from this conservation requirement over all cells, rather than guessed from this number alone.

Let p_{corr} denote the pressure sought by this correction solve, and q_{corr} its corrected velocity. Define their increments by

$$p' = p_{\text{corr}} - p^k, \qquad q' = q_{\text{corr}} - q^*.$$

A prime denotes an additive change, not a derivative. Equivalently,

$$\boxed{\Delta p = p_{\text{corr}} - p^k = p'.}$$

This is a pressure change at a fixed cell between estimates. The difference $p_E - p_P$ compares pressures at two locations in one estimate. The residual $r_{c,P}^*$ instead measures a velocity imbalance. These three quantities have different roles and must not be interchanged.

5 continued Derive the velocity correction from one momentum row

For horizontal face e, the predictor equation is

$$a_e u_e^* = \sum_{g \in \mathrm{NB}(e)} a_{eg} q_g^* + b_e - \frac{p_E^k - p_P^k}{h}.$$

Use the same coefficients for a corrected solution. Substitute $u_{e,\mathrm{corr}} = u_e^* + u_e'$, $q_{g,\mathrm{corr}} = q_g^* + q_g'$ and $p_{P,\mathrm{corr}} = p_P^k + p_P'$, with the analogous expression at E:

$$a_e(u_e^* + u_e') = \sum_{g \in \mathrm{NB}(e)} a_{eg}(q_g^* + q_g') + b_e - \frac{(p_E^k + p_E') - (p_P^k + p_P')}{h}.$$

Subtract the predictor row. The known source b_e, predictor neighbor terms and old pressure gradient cancel, leaving

$$a_e u_e' = \sum_{g \in \mathrm{NB}(e)} a_{eg} q_g' - \frac{p_E' - p_P'}{h}.$$

This exact correction equation couples corrections at neighboring velocity locations. For all velocity rows it is $Aq' = -Gp'$.

The SIMPLE approximation. For the local velocity correction, neglect the neighbor-correction sum. The center response remains:

$$a_e u_e' \simeq -\frac{p_E' - p_P'}{h}, \qquad \boxed{u_e' = -d_e \frac{p_E' - p_P'}{h}}, \qquad d_e = 1/a_e.$$

This approximates the momentum response to a pressure increment. It does not remove the requirement to conserve mass. The pressure equation is chosen so that the resulting face corrections satisfy every cell's continuity balance.

The same construction at the other three faces gives

$$u_w' = -d_w \frac{p_P' - p_W'}{h}, \qquad v_n' = -d_n \frac{p_N' - p_P'}{h}, \qquad v_s' = -d_s \frac{p_P' - p_S'}{h}.$$

Each d is the reciprocal diagonal of that face's own momentum row. Corrected continuity on a square cell requires

$$0 = r_{c,P}^* + \frac{u_e' - u_w' + v_n' - v_s'}{h}.$$

Section 6 substitutes the four expressions into this balance to obtain the pressure equation. Repeated coupling iterations also restore consistency with the full momentum equations, whose neighbor corrections were approximated.

5 continued One shared face, one velocity correction

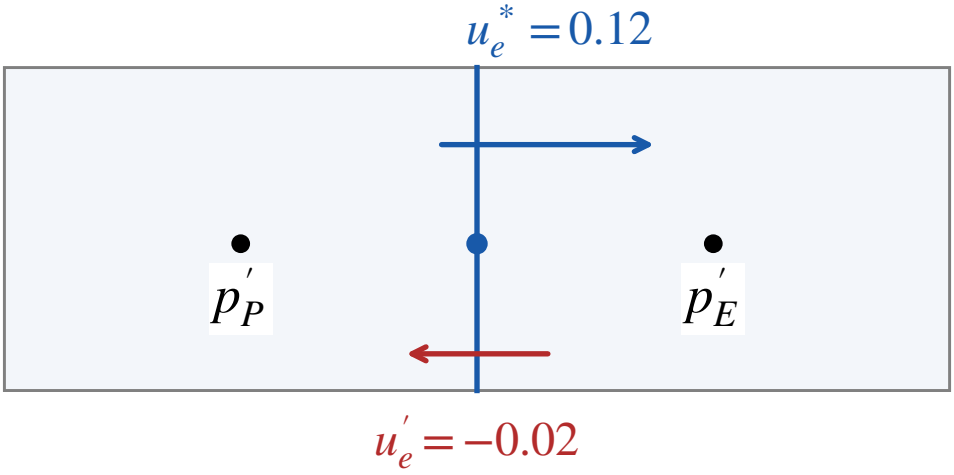


Figure 7: A shared-face pressure correction. When p'_E exceeds p'_P by 0.01, the velocity increment is westward: $u'_e = -0.02$. The predictor 0.12 becomes 0.10. The same face velocity enters the two neighboring outward flux balances with opposite signs.

The predictor u_e^* is obtained from momentum with the current pressure estimate. It generally leaves a cell mass imbalance. A pressure correction p' produces a velocity increment u'_e:

$$u_e = u_e^* + u'_e, \qquad u'_e = -d_e \frac{p'_E - p'_P}{h}, \qquad d_e = 1/a_e > 0.$$

Here a_e is the diagonal of the *volume-normalized* momentum equation. The prime denotes an increment, not differentiation. If $p'_E > p'_P$, the correction acts westward, consistent with acceleration down a pressure gradient.

Cell P and cell E use the same shared face value. In the outward flux balance it contributes $+hu_e$ for P and $-hu_e$ for E. A shared-face correction therefore preserves pairwise cancellation between neighboring cells. The pressure equation chooses these corrections to remove each cell's predictor mass imbalance, subject to the boundary fluxes and pressure reference.

Return to the predictor mass-balance example in Section 5, where $u_e^* = 0.12$ and $r_{c,P}^* = 0.08$. For this correction illustration, assume $a_e = 2$, so $d_e = 0.5$, and suppose the pressure solve returns $p'_E - p'_P = 0.01$. Then

$$u'_e = -0.5(0.01)/0.25 = -0.02, \qquad u_e = 0.12 - 0.02 = 0.10.$$

Holding the other faces fixed, cell P loses $h(0.02) = 0.005$ of outward volume flux: its imbalance becomes zero. The same face change increases cell E's outward flux by 0.005. A correction that fixes P alone may therefore spoil E; all pressure cells must be solved together. The prescribed diagonal $a_e = 2$ belongs to this correction illustration, not to the earlier convection-flux example.

6 The cell pressure equation and its boundary data

At each face, the corrected velocity is tentative velocity plus its correction:

$$u_e = u_e^* - \frac{d_e}{h}(p'_E - p'_P), \qquad u_w = u_w^* - \frac{d_w}{h}(p'_P - p'_W),$$
$$v_n = v_n^* - \frac{d_n}{h}(p'_N - p'_P), \qquad v_s = v_s^* - \frac{d_s}{h}(p'_P - p'_S).$$

Insert these into $(u_e - u_w + v_n - v_s)/h = 0$. The west and south terms change sign because their velocities enter cell outflow with minus signs:

$$0 = r^*_{c,P} - \frac{d_e}{h^2}(p'_E - p'_P) + \frac{d_w}{h^2}(p'_P - p'_W) - \frac{d_n}{h^2}(p'_N - p'_P) + \frac{d_s}{h^2}(p'_P - p'_S).$$

Collect the coefficients of each pressure correction and move the tentative residual to the right:

$$\frac{d_e + d_w + d_n + d_s}{h^2} p'_P - \frac{d_e p'_E + d_w p'_W + d_n p'_N + d_s p'_S}{h^2} = -r^*_{c,P}.$$

The center coefficient is the sum of four face weights; each neighbor has a negative coefficient. The right side is the negative cell divergence of the tentative velocity. The same face weight must be shared by its two adjacent pressure cells.

Matrix form. Name the resulting pressure matrix L. It has one row per pressure cell and multiplies the column of cell pressure corrections. This operator symbol is distinct from the cavity length used in Section 1's dimensional scaling. Applying D to $q = q^* - dGp'$ gives

$$0 = Dq^* - DdGp',$$
$$L = -DdG,$$
$$Lp' = -Dq^*.$$

The positive-diagonal convention for L explains the minus sign on the right. Once pressure is solved, the face correction $-dGp'$ cancels the computed cell imbalance, up to linear-solve error.

6 continued Boundary cells and pressure reference

Impermeable walls. The normal wall velocity is prescribed and is not an unknown interior face velocity. Its normal correction is zero. A prescribed wall-normal face contributes zero correction weight, so that face coefficient is omitted from the boundary-cell pressure row. Pressure assembly therefore has no correction flux across the external wall. For a cell adjacent to a stationary west wall, $u^*_w = u'_w = 0$, so its row is

$$\frac{d_e + d_n + d_s}{h^2} p'_P - \frac{d_e p'_E + d_n p'_N + d_s p'_S}{h^2} = -r^*_{c,P}.$$

There is no west pressure-neighbor term. At a corner, omit both prescribed wall-normal face weights. Zero normal correction does not mean zero pressure at every wall.

Pressure reference. The vector $\mathbf{1}$ contains one in every pressure entry. Adding a constant to every cell pressure changes no pressure gradient: $G\mathbf{1} = 0$. Consequently $L\mathbf{1} = 0$ and the closed-cavity pressure system has one arbitrary constant. Pinning means temporarily prescribing one pressure unknown, for example $p_{\text{ref cell}} = 0$, to make the linear solve unique. The implementation then removes the arithmetic cell mean:

$$\bar{p} = \frac{1}{N^2} \sum_P p_P, \qquad \widetilde{p}_P = p_P - \bar{p}.$$

The same constant is subtracted from every cell, so

$$\widetilde{p}_E - \widetilde{p}_P = (p_E - \bar{p}) - (p_P - \bar{p}) = p_E - p_P.$$

The gradient and corrected velocity are unchanged. The reported zero-mean pressure is a gauge choice, not an extra boundary force.

Tangential no-slip is enforced in momentum through wall diffusion. On the staggered grid it is distinct from the zero normal correction used in the cell mass balance.

7 Two-cell pressure-correction example

Consider two closed square cells with one shared vertical face. This is an algebraic conservation example, not a resolved cavity. The pressure guess is initially zero in both cells. Let $h = 0.5$, tentative shared velocity $u^* = 0.2$, and inverse momentum diagonal $d_e = 0.1$. Positive shared velocity leaves cell 1 and enters cell 2. All external normal velocities are zero.

$$r^*_{c,1} = \frac{0.2}{0.5} = 0.4, \qquad r^*_{c,2} = \frac{-0.2}{0.5} = -0.4.$$

The shared pressure weight is $d_e/h^2 = 0.1/0.25 = 0.4$. Therefore

$$0.4 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} p'_1 \\ p'_2 \end{bmatrix} = \begin{bmatrix} -0.4 \\ 0.4 \end{bmatrix}.$$

The first row gives $p'_1 - p'_2 = -1$. With zero mean, $p'_1 + p'_2 = 0$, so $p'_1 = -0.5$ and $p'_2 = 0.5$. The corrected face velocity is

$$u = u^* - d_e \frac{p'_2 - p'_1}{h} = 0.2 - 0.1 \frac{0.5 - (-0.5)}{0.5} = 0.$$

Thus the unrelaxed changes are $\Delta p_1 = p'_1 = -0.5$ and $\Delta p_2 = p'_2 = 0.5$. Both corrected divergences are zero. A common constant added to p'_1 and p'_2 leaves this calculation unchanged.

Pressure relaxation and velocity correction. The pressure relaxation factor α_p is the fraction of p' added to the pressure stored for the next momentum predictor: $p^{k+1} = p^k + \alpha_p p'$. The stored pressure change is therefore $\alpha_p p'$, rather than the full unrelaxed $\Delta p = p'$. If only $\alpha_p = 0.3$ of the velocity correction were applied, the shared velocity would remain $0.2 - 0.3(0.2) = 0.14$. The two mass defects would then be $+0.28$ and -0.28. With $\alpha_p = 0.3$, the stored pressures become -0.15 and 0.15 from the zero initial guess, while the fully corrected shared velocity remains zero. This velocity need not yet satisfy momentum with the relaxed stored pressure; the next predictor and correction continue the coupling iteration. The conservative velocity correction is applied fully.

Local and global conservation. The initial global sum was already $0.4 - 0.4 = 0$, but each cell was imbalanced. A zero total boundary flux alone cannot certify local continuity.

7 continued Prepare a steady iteration: under-relaxation

A coupling *iteration* updates an estimate of a steady solution. The counter k labels those updates. It does not measure elapsed physical time. The estimates entering the update are q^k and p^k.

Velocity under-relaxation. Convection coefficients depend on velocity, so replacing an estimate by a very different predictor can destabilize the next coefficient update. An under-relaxation factor α_u, with $0 < \alpha_u \leq 1$, controls this change. Subscript u refers to velocity; the same factor is used for both components here.

Let a_f^0 be the center coefficient before relaxation. The superscript 0 identifies the original coefficient of this assembly, not an initial time. The relaxed row uses

$$a_f^{\text{rel}} = \frac{a_f^0}{\alpha_u}, \qquad b_f^{\text{rel}} = b_f + \left(\frac{1}{\alpha_u} - 1\right) a_f^0 q_f^k.$$

The label rel means relaxed. Neighbor coefficients are unchanged. The larger diagonal is accompanied by a known source containing the old velocity estimate. At a converged fixed point the new and old velocities coincide, so these extra diagonal and source terms cancel. The intended steady momentum equation is preserved.

For the preceding single-row example, let $a_e^0 = 0.64$, the old estimate be $u_e^k = 0.02$, and choose $\alpha_u = 0.7$. Holding its neighbor and pressure contributions fixed gives

$$\begin{aligned} a_e^{\text{rel}} &= 0.64/0.7 \simeq 0.9142857, \\ \text{added source} &= (1/0.7 - 1)(0.64)(0.02) \simeq 0.005485714, \\ u_e^* &\simeq (0.0152 + 0.008 + 0.005485714)/0.9142857 = 0.031375. \end{aligned}$$

For this frozen-row illustration, this is also $0.7(0.03625) + 0.3(0.02)$. In the coupled solver, relaxation is applied to the assembled momentum equations as written above. The pressure correction uses the relaxed reciprocal diagonal $d_f = 1/a_f^{\text{rel}}$.

Pressure under-relaxation. The separate factor α_p, with $0 < \alpha_p \leq 1$, controls the pressure stored for the next predictor: $p^{k+1} = p^k + \alpha_p p'$. The full velocity correction is retained to satisfy continuity, as the two-cell example demonstrated. The factors α_u and α_p are numerical controls, not fluid properties.

7 continued Decide whether both equations have been satisfied

A *residual* is the left side of an equation minus its right side, evaluated using the current estimates. Zero residual means that discrete equation is satisfied. The cell continuity residual is already $r_{c,P} = (Dq)_P$.

For the momentum equations, collect their residuals into a column r_m:

$$r_m = A(q)q - b + Gp.$$

Here $A(q)$ means that convection coefficients are rebuilt from the current velocity before evaluating this test. Use the original, unrelaxed steady equations. This checks the physical discretization being solved rather than only the temporary relaxed predictor equations.

To report one number for a residual column r, use the infinity norm

$$\|r\|_\infty = \max_i |r_i|.$$

Index i runs through that column's entries; the operation selects the largest absolute entry. Thus the two required checks are

$$\|r_m\|_\infty < \varepsilon_m, \qquad \|Dq\|_\infty < \varepsilon_c.$$

The positive numbers ε_m and ε_c are the selected momentum and continuity tolerances. They are separate because the residuals test different equations. The retained cavity runs use 10^{-6} and 10^{-9} respectively.

A small change between successive velocities is insufficient: strong under-relaxation can produce a small change even when the momentum balance is still inaccurate. Small equation residuals are also distinct from small spatial discretization error. Accuracy still requires the reference-profile and mesh-refinement comparisons later in the lesson.

In a steady coupling iteration, both equation checks are evaluated after the velocity and pressure updates, using the latest corrected fields.

8 SIMPLE: assemble the steady coupling iteration

SIMPLE means Semi-Implicit Method for Pressure-Linked Equations. It combines the momentum solve, pressure correction and conservative velocity update already derived. Begin iteration k with velocity q^k and pressure p^k.

1. **Build momentum.** Evaluate convection from q^k and assemble the original steady matrix A and known source b. Apply the velocity under-relaxation rule from Section 7 to obtain A^{rel} and b^{rel}. The relaxed center coefficients become the diagonal of A^{rel}; its neighbor entries are unchanged. The column b^{rel} collects the relaxed source entries.
2. **Predict velocity.** Solve the relaxed momentum equations using the current pressure. The result is q^*, which may violate continuity.
3. **Calculate the imbalance.** Apply D to obtain one residual $r^*_{c,P}$ per pressure cell. Use the relaxed inverse diagonal d to build the pressure matrix $L = -DdG$.
4. **Solve pressure correction.** Solve $Lp' = -Dq^*$ with the wall correction conditions and one pressure reference, as derived in Section 6.
5. **Update the fields.** Apply the full face-velocity correction and the under-relaxed pressure update shown below.
6. **Test both equations.** Rebuild the unrelaxed momentum coefficients from the corrected velocity and check the momentum and continuity tolerances. If either fails, use the corrected fields as the next estimates and repeat.

In compact notation, the solve and updates are

$$\begin{aligned} A^{\text{rel}} q^* &= b^{\text{rel}} - Gp^k, \\ Lp' &= -Dq^*, \\ q^{k+1} &= q^* - dGp', \\ p^{k+1} &= p^k + \alpha_p p'. \end{aligned}$$

The stored pressure increment is $\alpha_p p'$, whereas the unrelaxed correction is $\Delta p = p'$. Velocity receives its full correction so that the cell balances are conservative. Prescribed wall velocities remain fixed; a common pressure offset is removed without changing pressure gradients.

The retained baseline uses $\alpha_u = 0.7$, $\alpha_p = 0.3$, $\varepsilon_m = 10^{-6}$ and $\varepsilon_c = 10^{-9}$. These numerical controls do not change the Reynolds number. This SIMPLE branch solves a steady problem; increasing k does not advance physical time.

8 continued SIMPLE: follow the loop

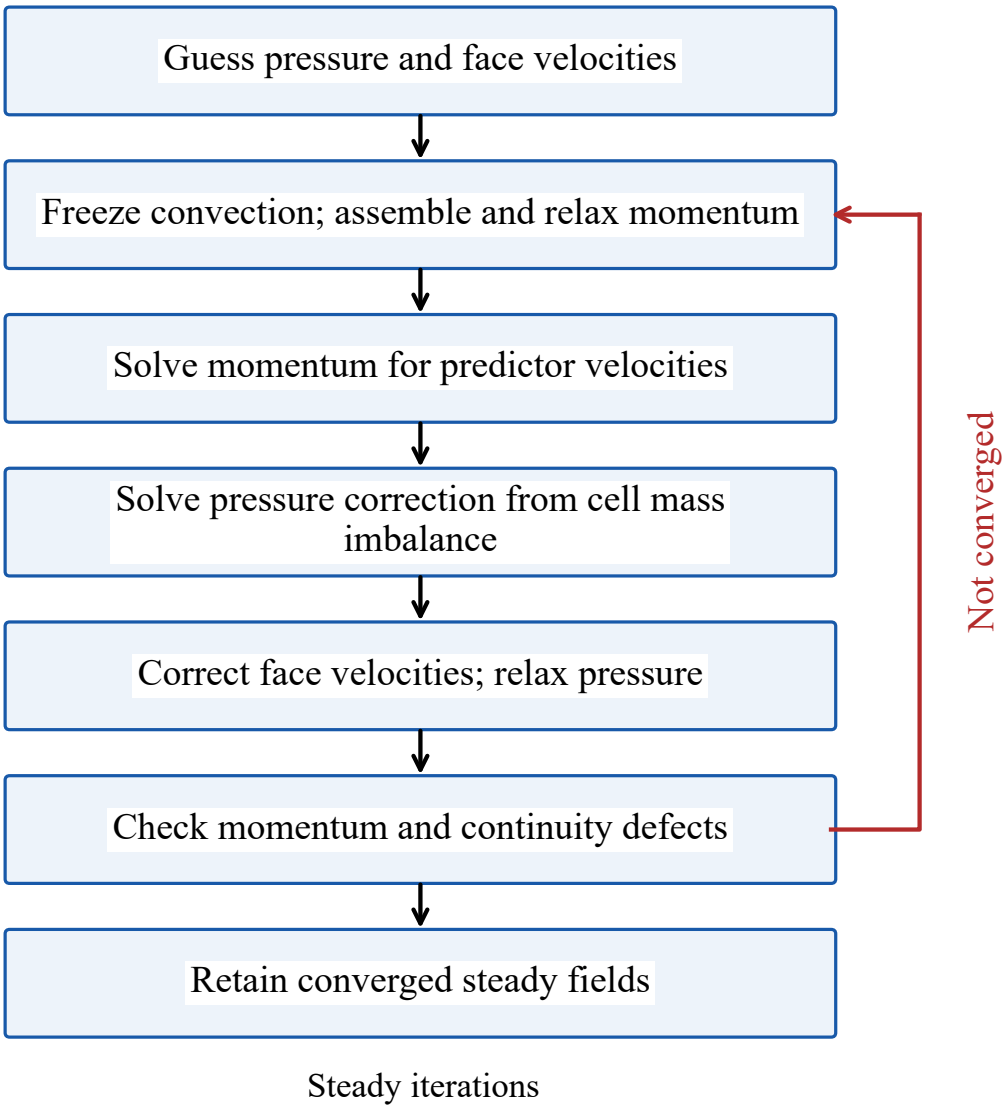


Figure 8: SIMPLE for a steady problem. Follow the blue boxes downward. The red return path rebuilds momentum after an unconverged iteration. Correct face velocities fully, relax pressure as specified, and test both momentum and cell continuity.

A return along the red arrow begins a new steady nonlinear iteration. Rebuild the convection coefficients from the latest velocity and check both momentum and continuity.

9 Physical time and backward-Euler momentum

PISO and PIMPLE solve a transient momentum equation. Superscripts n and $n+1$ denote physical time levels, separated by Δt. For one face velocity,

$$\left.\frac{\partial q_f}{\partial t}\right|_{n+1} \approx \frac{q_f^{n+1} - q_f^n}{\Delta t}.$$

If the spatial momentum row has center coefficient a_f^{space} and source b_f^{space}, backward Euler produces

$$\left(a_f^{\text{space}} + \frac{1}{\Delta t}\right) q_f^{n+1} = \sum_g a_{fg} q_g^{n+1} + b_f^{\text{space}} + \frac{q_f^n}{\Delta t} - (Gp^{n+1})_f.$$

Thus the transient assembly adds $1/\Delta t$ to the diagonal and the known old velocity $q_f^n/\Delta t$ to the source.

Old-time velocity is fixed. At the beginning of the physical step, store q^n. Every momentum assembly and every correction within that step uses the same old-time source. Replacing q^n with an intermediate corrected velocity would change the discrete physical problem.

Nonlinear coefficients. Convective coefficients depend on advecting velocity. They must be evaluated from a known estimate to form a linear predictor system. Such coefficients are called frozen or lagged during that solve. A perfectly solved linear predictor can still differ from the fully implicit nonlinear step if its coefficients were evaluated at an outdated velocity.

Three iteration levels. A linear solver solves one matrix equation. A pressure corrector couples pressure and velocity for a frozen momentum matrix. An outer corrector rebuilds that matrix using the latest velocity. None of these is an additional physical time advance.

Backward Euler is first-order accurate in time. Extra correctors can reduce algebraic coupling error, but do not change its time-discretization order. A larger stable Δt still requires a temporal-accuracy assessment.

10 PISO: several corrections in one physical time step

PISO means Pressure-Implicit with Splitting of Operators. Start with the accepted old-time velocity q^n and pressure p^n. The next accepted fields will belong to time $n+1$, one physical interval Δt later.

First assemble the transient momentum equation from Section 9 and solve

$$Aq^* = b - Gp^{\text{guess}}.$$

The pressure p^{guess} is the initial pressure estimate for this step, normally the preceding accepted pressure. The source b includes the fixed old-time contribution $q^n/\Delta t$ and known wall contributions. The predictor q^* satisfies this momentum solve, but can still violate continuity. Convection coefficients are evaluated from the initial velocity estimate, for example q^n, and held fixed during the following inner corrections.

The next operations improve pressure and velocity within this same step. Write $q^{(0)} = q^*$ and use m to count inner corrections. Parenthesized superscripts identify a correction estimate, not a physical time or a power.

Quantity	Behavior during this PISO loop
q^n	Accepted old-time velocity; fixed throughout the step.
A, b	Transient momentum coefficients and source; assembled once.
d, R	Reciprocal diagonal and off-diagonal part of this same A; fixed.
$L = -DdG$	Pressure matrix from the fixed reciprocal diagonal; fixed.
$q^{(m)}, p^{(m)}$	Current estimates at the new time level; updated by corrections.
$H^{(m)}$	Known source plus the latest neighbor-velocity contributions; recomputed.

The key distinction is between a fixed momentum matrix and changing velocity values within it. Updating the neighbor contributions is what makes another pressure correction different from solving an identical equation again.

10 continued Form each inner pressure solve and velocity update

Given the current estimate $q^{(m)}$, collect its neighbor contributions:

$$H^{(m)} = b - Rq^{(m)}.$$

The momentum-row reconstruction for a new pressure estimate is

$$q^{(m+1)} = d[H^{(m)} - Gp^{(m+1)}].$$

Require its cell divergence to vanish. Applying D gives

$$\begin{aligned} 0 &= D(dH^{(m)}) - DdGp^{(m+1)}, \\ Lp^{(m+1)} &= -D(dH^{(m)}), \qquad L = -DdG. \end{aligned}$$

Thus one inner correction consists of recomputing $H^{(m)}$, solving this pressure equation and reconstructing all face velocities. The next correction repeats these operations with the newly reconstructed velocity.

Full pressure or increment? In this implementation, $p^{(m+1)}$ is the full pressure field in the selected reference convention. It replaces the previous pressure estimate. It is not SIMPLE's additive correction p', and must not be added to the previous pressure.

Let N_i denote the chosen number of inner corrections. After those corrections, accept $q^{n+1} = q^{(N_i)}$ and $p^{n+1} = p^{(N_i)}$. Only this acceptance advances the physical time. The retained baseline uses $N_i = 2$ and does not apply SIMPLE's under-relaxation in this transient branch.

The convection coefficients remain those assembled for the predictor. Further inner corrections improve coupling for that fixed matrix. They do not rebuild convection from the latest new-time velocity; PIMPLE adds an outer iteration for that purpose.

10 continued PISO: follow the loop

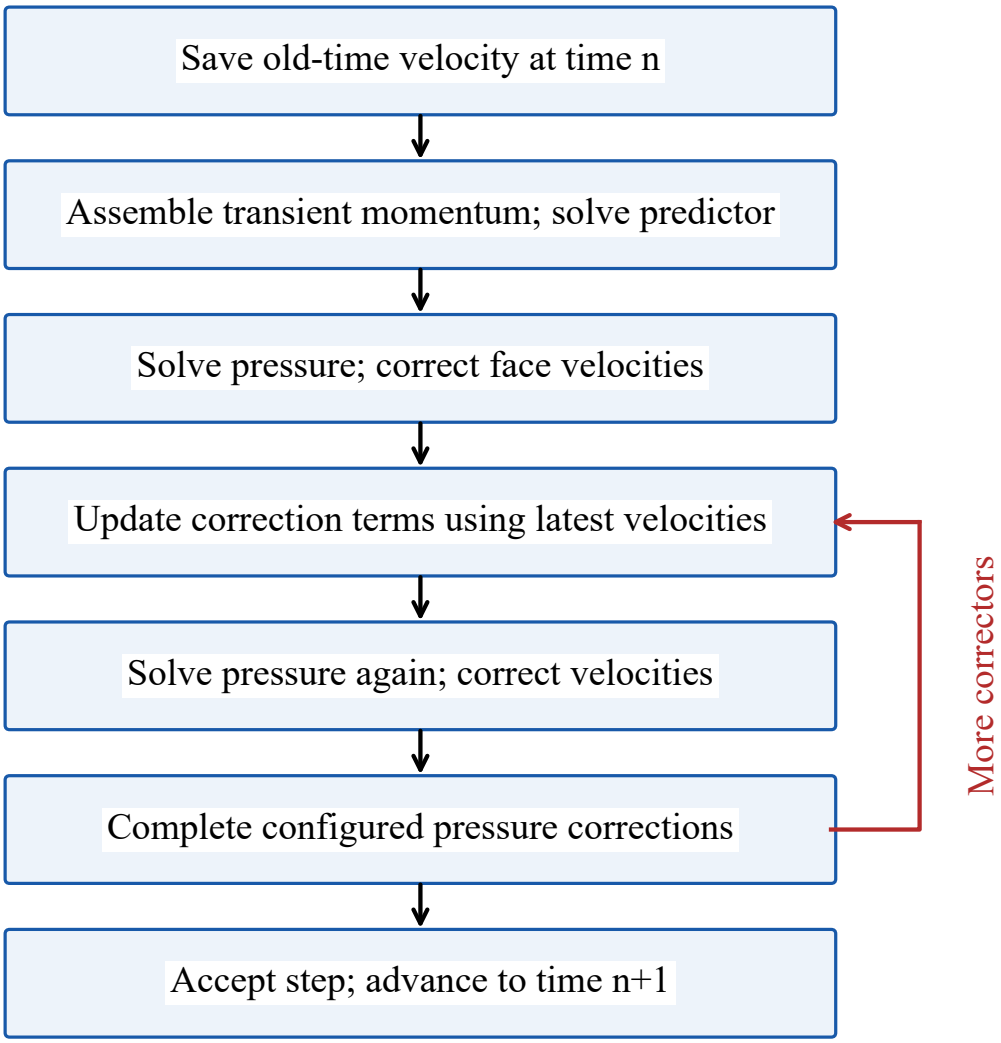


Figure 9: PISO within one physical time step. The red return path updates correction terms before another pressure solve, using fixed momentum coefficients. The saved old-time velocity remains unchanged until the new time level is accepted.

All inner corrections remain at the same new time level. The old-time velocity in the transient term is fixed. Each further pressure correction uses updated velocity information; an unchanged repeated pressure solve is not an additional PISO correction.

11 PISO: a frozen-matrix numerical example

This reduced two-velocity system illustrates coupling algebra; it is not a cavity grid. Let

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & -1 \end{bmatrix}, \quad G = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Here p is one independent scalar pressure variable after removing an arbitrary common pressure offset. It is not a separate pressure value in each physical cell. The matrices are prescribed to illustrate coupling algebra; they are not derived from the earlier two-cell geometry. Continuity is $Dq = q_1 - q_2 = 0$. From A,

$$d = \begin{bmatrix} 1/4 & 0 \\ 0 & 1/3 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad L = -DdG = 7/12.$$

Starting from $p^{\text{guess}} = 0$, the momentum predictor solves $Aq^* = b$:

$$q^* = \begin{bmatrix} 6/11 \\ -2/11 \end{bmatrix}, \qquad Dq^* = 8/11.$$

It satisfies momentum exactly but violates continuity. The first inner correction uses

$$H^{(0)} = b - Rq^* = \begin{bmatrix} 24/11 \\ -6/11 \end{bmatrix}, \qquad dH^{(0)} = \begin{bmatrix} 6/11 \\ -2/11 \end{bmatrix}.$$

Solve pressure and reconstruct velocity:

$$\frac{7}{12}p^{(1)} = -D(dH^{(0)}) = -\frac{8}{11}, \qquad p^{(1)} = -\frac{96}{77},$$
$$q_1^{(1)} = \frac{1}{4}\left(\frac{24}{11} + p^{(1)}\right) = \frac{18}{77}, \qquad q_2^{(1)} = \frac{1}{3}\left(-\frac{6}{11} - p^{(1)}\right) = \frac{18}{77}.$$

The signs in these two velocity formulas follow from $G = (-1, 1)^T$. Now $Dq^{(1)} = 0$. Nevertheless, substituting these fields into the original frozen momentum equation gives

$$r^{(1)} = Aq^{(1)} - b + Gp^{(1)} = \begin{bmatrix} 32/77 \\ -24/77 \end{bmatrix}.$$

The vector $r = Aq - b + Gp$ is the error in satisfying the momentum rows; exact satisfaction gives $r = 0$. Its infinity norm is the largest absolute entry, $\|r\|_\infty = \max_i |r_i|$, where i labels a momentum row. Mass balance is exact here, but momentum consistency still needs improvement.

11 continued The second pressure correction

The corrected neighbor velocities change H, even though A and L are unchanged:

$$H^{(1)} = b - Rq^{(1)} = \begin{bmatrix} 136/77 \\ -18/77 \end{bmatrix}, \qquad dH^{(1)} = \begin{bmatrix} 34/77 \\ -6/77 \end{bmatrix}.$$

The next pressure and velocity are

$$\frac{7}{12}p^{(2)} = -\frac{40}{77}, \qquad p^{(2)} = -\frac{480}{539},$$
$$q^{(2)} = d\left[H^{(1)} - Gp^{(2)}\right] = \begin{bmatrix} 118/539 \\ 118/539 \end{bmatrix}.$$

The pressure is replaced by $p^{(2)}$; adding $p^{(2)}$ to $p^{(1)}$ would be incorrect. Again continuity is exact. The remaining frozen-momentum defect is

$$r^{(2)} = \begin{bmatrix} -8/539 \\ -8/539 \end{bmatrix}.$$

Stage	$\lvert Dq\rvert$	$\lVert Aq - b + Gp\rVert_\infty$
Momentum predictor	$8/11$	0
Inner correction 1	0	$32/77 \approx 0.41558$
Inner correction 2	0	$8/539 \approx 0.01484$

The exact constrained solution is $q_1 = q_2 = 2/9$ and $p = -8/9$. It follows by imposing $q_1 = q_2 = s$ in both original momentum rows: $5s - p = 2$ and $4s + p = 0$, hence $9s = 2$.

The second correction reduces momentum inconsistency in this example while preserving mass balance. This is a demonstrated property of these specified matrices, not a guarantee of monotone convergence for every CFD system.

The right-hand side of the pressure equation changes from $-8/11$ to $-40/77$. The example therefore distinguishes another PISO correction from another iteration of an unchanged linear pressure solve.

12 PIMPLE: outer coefficient updates and inner corrections

PIMPLE combines an outer momentum-update loop with PISO inner corrections. Physical time levels are $n, n+1$; ℓ labels outer updates and m labels inner pressure corrections. The previous physical velocity q^n is stored once.

Before writing the nested equations, distinguish their labels and stored fields:

Label or field	Meaning
$n, n+1$	Old and new physical time levels.
ℓ	Outer pass: rebuild momentum coefficients using the latest velocity estimate.
m	Inner correction within one outer pass: update pressure and face velocity.
q^n	Accepted old-time velocity, saved once and kept fixed.
$q_{\text{lag}}^{(\ell)}, p_{\text{lag}}^{(\ell)}$	Velocity and pressure estimates entering outer pass ℓ. The subscript lag identifies inputs carried from the preceding pass.
$q^{(\ell,0)}$	Momentum predictor at the start of outer pass ℓ.
$q^{(\ell,m)}$	New-time velocity estimate after m inner corrections in pass ℓ.

For example, $(\ell, m) = (2, 1)$ means the first inner correction of the second outer pass. It remains within the same physical step. The first outer pass starts from the accepted old-time fields; subsequent passes start from the latest corrected new-time fields.

12 continued Assemble an outer pass and carry its fields forward

At the start of outer pass ℓ, assemble convection using the latest available corrected velocity $q_{\text{lag}}^{(\ell)}$. Here A_{space} contains the spatial convection and diffusion coefficients, b_{wall} contains prescribed wall contributions, and I is the identity matrix (ones on its diagonal, zeros elsewhere). Adding the backward-Euler terms gives

$$A^{(\ell)} = A_{\text{space}}(q_{\text{lag}}^{(\ell)}) + I/\Delta t, \qquad b^{(\ell)} = b_{\text{wall}}^{(\ell)} + q^n/\Delta t.$$

The same q^n appears in every outer pass. For this fixed-grid, fixed-lid code, the wall source is constant, so $b^{(\ell)}$ has the same value in every outer pass. The prescribed wall motion and old-time source belong to the same physical step.

The first outer pass starts from q^n, p^n. The symbol $p_{\text{lag}}^{(\ell)}$ denotes the latest available pressure at the start of outer pass ℓ. A new momentum predictor is solved using that pressure. The predictor is therefore

$$A^{(\ell)} q^{(\ell,0)} = b^{(\ell)} - G p_{\text{lag}}^{(\ell)}.$$

12 continued Inner corrections and the next outer pass

From this assembly define $d^{(\ell)}$ as its reciprocal diagonal, $R^{(\ell)}$ as its off-diagonal part, and $L^{(\ell)} = -Dd^{(\ell)}G$ as its pressure matrix. These three objects are fixed during the inner corrections of outer pass ℓ:

$$\begin{aligned} H^{(\ell,m)} &= b^{(\ell)} - R^{(\ell)} q^{(\ell,m)}, \\ L^{(\ell)} p^{(\ell,m+1)} &= -D\left(d^{(\ell)} H^{(\ell,m)}\right), \\ q^{(\ell,m+1)} &= d^{(\ell)}\left[H^{(\ell,m)} - G p^{(\ell,m+1)}\right]. \end{aligned}$$

With N_i inner corrections per outer pass, the initial and transfer rules are

$$\begin{aligned} q_{\text{lag}}^{(1)} &= q^n, & p_{\text{lag}}^{(1)} &= p^n, \\ q_{\text{lag}}^{(\ell+1)} &= q^{(\ell,N_i)}, & p_{\text{lag}}^{(\ell+1)} &= p^{(\ell,N_i)}. \end{aligned}$$

Thus the final corrected fields of one outer pass initialize the next. Rebuilding A also rebuilds d, R and L. Each inner solve computes the full pressure field in the chosen gauge, as in this PISO implementation.

Two different purposes. Inner corrections improve pressure–velocity consistency for the current matrix. Outer passes update the nonlinear coefficients that matrix represents. Neither loop advances physical time. Only after the final outer pass are the accepted fields assigned to time $n + 1$.

With one outer pass, this code reduces to its PISO algorithm. Its PIMPLE branch does not apply SIMPLE's under-relaxation. The retained code performs the configured number of outer and inner corrections, then tests steady convergence between completed steps. A transient-accuracy study must also assess the time step and remaining within-step defects. Additional loops reduce certain algebraic errors at extra cost; they do not increase backward-Euler's formal time order.

12 continued PIMPLE: follow the loop

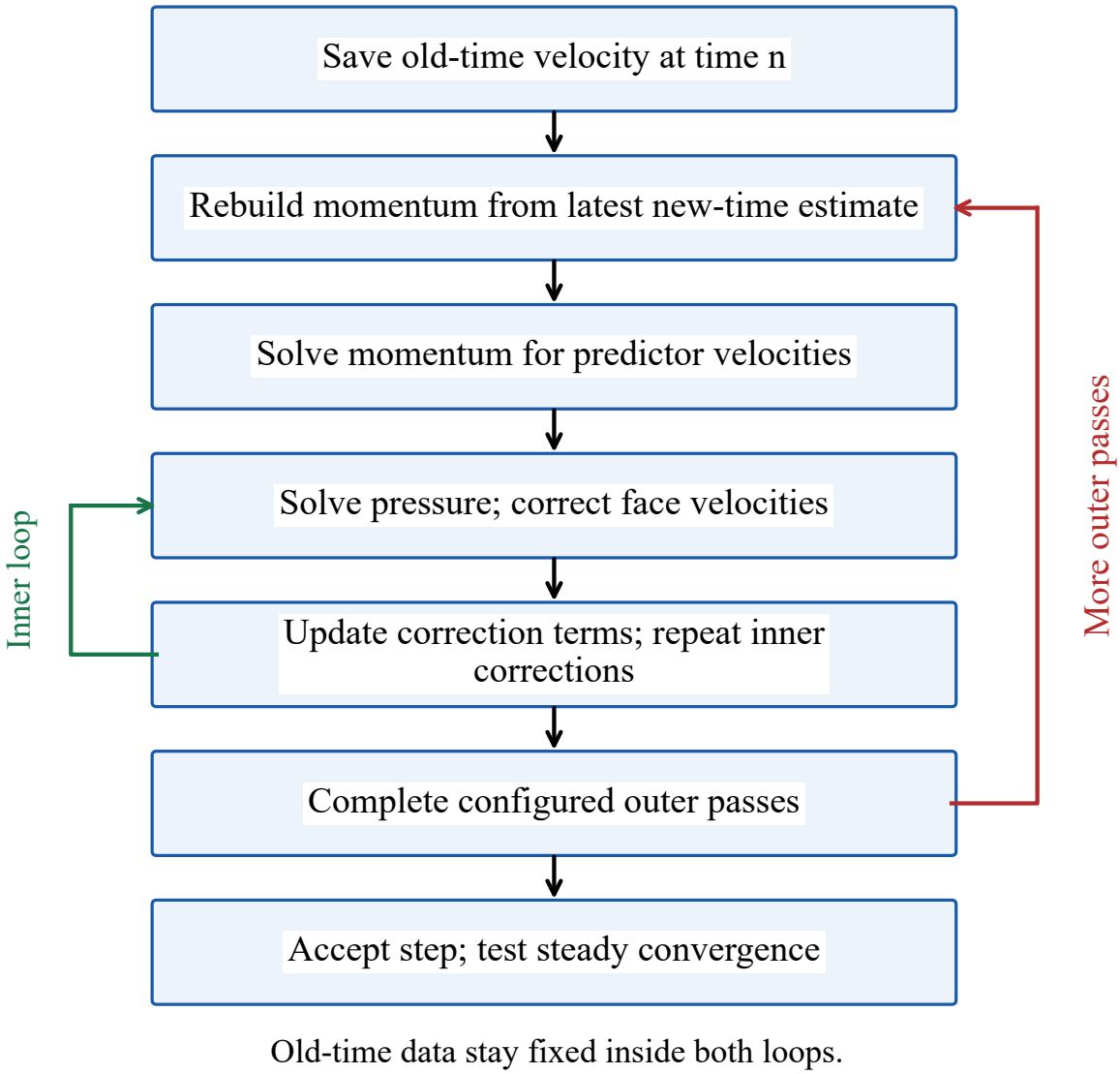


Figure 10: PIMPLE within one physical time step. Green: inner pressure-velocity corrections with fixed momentum coefficients. Red: outer passes rebuild those coefficients from the latest new-time estimate. The retained code completes the configured passes, accepts the step, and tests steady convergence between steps. Neither correction loop advances the saved old-time data.

The outer loop updates convection coefficients and resolves momentum. The inner loop updates pressure and face velocity for those coefficients. Neither loop advances physical time. The retained code completes the configured passes and checks steady convergence between steps. A time-accurate simulation also needs within-step and time-step accuracy checks.

13 PIMPLE: a nonlinear coefficient example

Extend the preceding algebraic example by making its momentum coefficients depend on velocity:

$$A(s) = \begin{bmatrix} 4+s & 1 \\ 1 & 3+s \end{bmatrix}, \qquad s = (q_1 + q_2)/2, \qquad b = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Keep the same D, G and initial $q = 0, p = 0$. At each outer pass, evaluate s from its starting velocity, freeze $A(s)$, solve a new predictor, and perform two inner corrections. The source b remains fixed. This toy coefficient law illustrates lagging; it is not a Navier–Stokes convection discretization.

At outer pass 1, $s = 0$, so the two inner corrections are exactly the previous PISO example. They end at $q_1 = q_2 = 0.2189239332$. The next outer pass uses

$$A^{(2)} = \begin{bmatrix} 4.2189239332 & 1 \\ 1 & 3.2189239332 \end{bmatrix},$$

and recomputes d, R, L before the new predictor and inner corrections.

Outer pass	Frozen s	Final $q_1 = q_2$	Final p	Nonlinear defect
1	0	0.2189239332	–0.8905380334	3.30854×10^{-2}
2	0.2189239332	0.2119026446	–0.8940486777	1.53537×10^{-3}
3	0.2119026446	0.2122289655	–0.8938855173	7.14784×10^{-5}

The reported defect is the largest absolute component of the nonlinear momentum residual, evaluated with coefficients rebuilt from the final velocity:

$$\| A\left((q_1 + q_2)/2\right) q - b + Gp \|_\infty .$$

It is different from the residual of the frozen predictor matrix. All final mass defects are below 10^{-15}.

For the exact constrained solution set $q_1 = q_2 = s$ and add the two momentum rows:

$$(9 + 2s)s = 2, \qquad s = \frac{\sqrt{97} - 9}{4} = 0.2122144504, \qquad p = s/2 - 1.$$

The table shows the benefit of coefficient updates for this example. Actual cavity efficiency must be judged using its own measured equation defects, accuracy and solve costs.

14 Algorithm loops and computational work

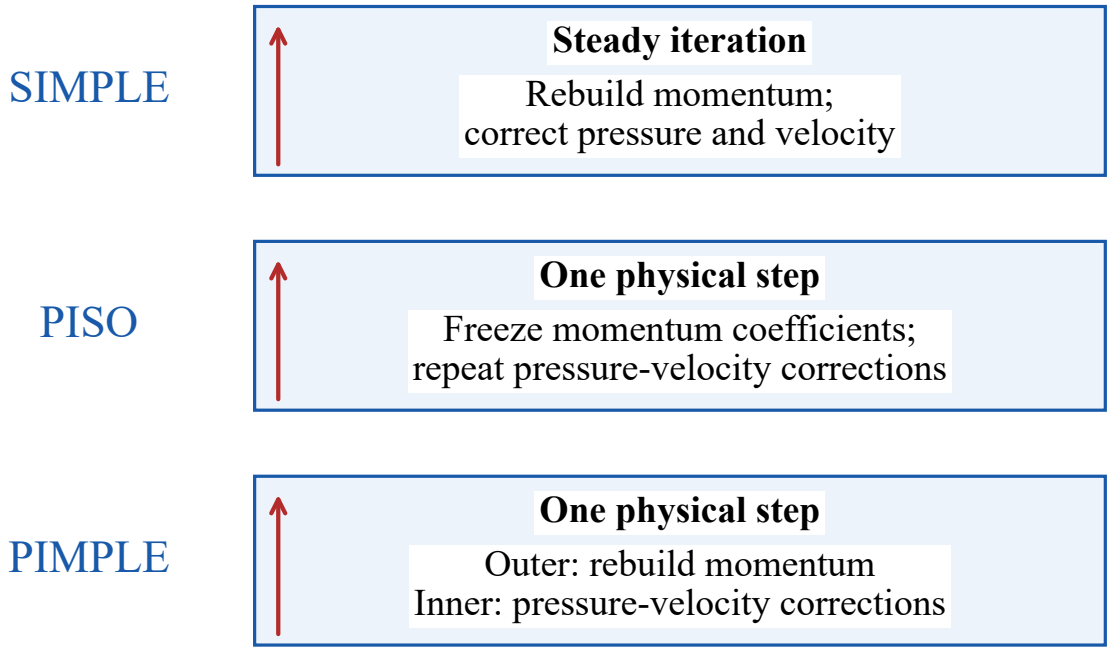


Figure 11: Comparison of repeated work. A SIMPLE cycle is a steady nonlinear iteration. PISO repeats pressure-velocity corrections within one step. PIMPLE nests those corrections inside momentum-coefficient updates. The red arrows denote repetition; they do not denote physical time advancement.

The nested loops act on one target problem at the current physical step. The diagram distinguishes steady SIMPLE iterations from PISO and PIMPLE time integration.

Let N_i be the number of pressure correctors and N_o the number of outer correctors. Each predictor solves horizontal and vertical momentum separately.

Work unit	SIMPLE iteration	PISO step	PIMPLE step
Momentum assemblies	1	1	N_o
Component momentum solves	2	2	$2N_o$
Pressure solves	1	N_i	N_oN_i
Physical time advances	0	1	1

For three outer and two inner correctors, PIMPLE uses three assemblies, six component momentum solves and six pressure solves per physical step. A direct linear solver may factor a pressure matrix once and reuse it within an inner loop, while solving different right-hand sides.

An iteration count alone cannot compare these costs. The retained runs use fixed corrector counts, rather than an adaptive inner tolerance. A steady stopping test is applied outside these operations.

15 Convergence, accuracy and the meaning of a residual

Cell continuity. The mass-balance diagnostic is $\max_P |r_{c,P}|$: the largest absolute continuity residual among all pressure cells. It is evaluated from face velocities, not from an interpolated picture of cell-center velocity.

Momentum defect. For a steady target, the implementation rebuilds the unrelaxed spatial coefficients using final velocity and evaluates the original momentum equations. For a transient step, a nonlinear step defect includes the same fixed old-time source:

$$r_{\text{step}} = A_{\text{space}}(q^{n+1})q^{n+1} - b_{\text{wall}} + \frac{q^{n+1} - q^n}{\Delta t} + Gp^{n+1}.$$

Small mass defect alone does not make this vector small, as the PISO numerical example demonstrated.

Accuracy. Agreement with independent reference velocities and stability under spatial refinement address solution accuracy. Ghia centerline tables validate selected velocity profiles; they do not validate pressure. A converged coarse-grid calculation can still be inaccurate.

Transient accuracy. Corrector loops reduce algebraic errors inside a step. Temporal error is assessed separately by reducing Δt and comparing at matched physical times. Steady cavity agreement alone does not validate startup or oscillatory flow.

Streamfunction–vorticity comparison. Week 1 advances vorticity and solves streamfunction. Its native update norm is different from the finite-volume momentum defect. Their convergence histories use separate axes; common velocity errors and wall-clock times provide cross-formulation comparisons. N finite-volume cells correspond to $N+1$ finite-difference nodes at matching spacing $h = 1/N$.

Higher Reynolds numbers. Stronger convection and thinner layers can require more cells and more work. More pressure corrections cannot recover spatial structures missing from the mesh. Refinement and higher-*Re* runs can take substantially longer and need more memory; the supplied coarse cases are the starting point for laptop or Colab use.

16 Recorded cavity accuracy and runtime at $Re = 100$

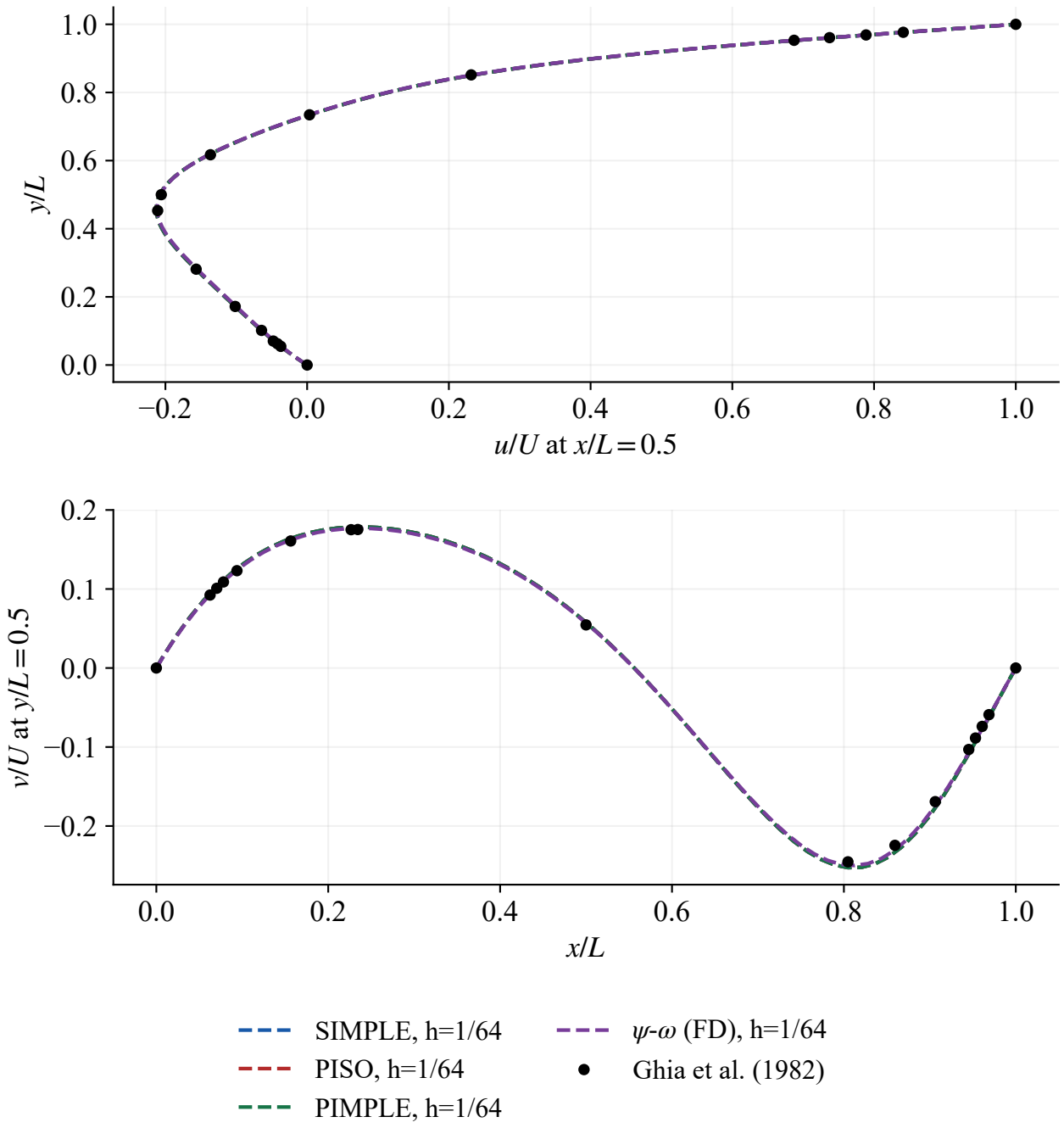


Figure 12: Retained Re=100 centerline velocities at matched spacing h=1/64. Curves show all four formulations; black markers are Ghia et al. (1982) samples. FV uses 64 pressure cells per side and FD uses 65 boundary-inclusive nodes. Line styles and colors identify the computed series.

The Ghia velocity markers are independent reference samples. The three finite-volume coupling methods use the same spatial equations. Their converged fields agree closely; differences from the finite-difference formulation also include spatial-discretization differences. The displayed 32/64 finite-volume cell counts correspond to 33/65 finite-difference nodes.

16 continued Accuracy and measured solver cost

Formulation	Cells per side	Iterations / steps	Recorded seconds
SIMPLE	32	480	5.074
PISO	32	540	6.994
PIMPLE	32	540	19.604
Streamfunction–vorticity	32	19000	8.106
SIMPLE	64	1870	137.987
PISO	64	540	42.756
PIMPLE	64	540	146.522
Streamfunction–vorticity	64	19000	17.284

The table includes all four formulations and reports recorded solver-call seconds. The finite-difference reference has its own stopping norm and recovers pressure afterward. These measurements depend on the host, implementation, mesh, relaxation and stopping targets; they do not establish a universal fastest method.

The finite-difference centerline errors decrease on refinement in this retained comparison. The finite-volume Ghia errors are not monotone between the two supplied grids. That behavior is retained rather than interpreted as proof of grid independence. More resolution and reference evidence are needed for an asymptotic accuracy claim.

17 Velocity structure and vorticity

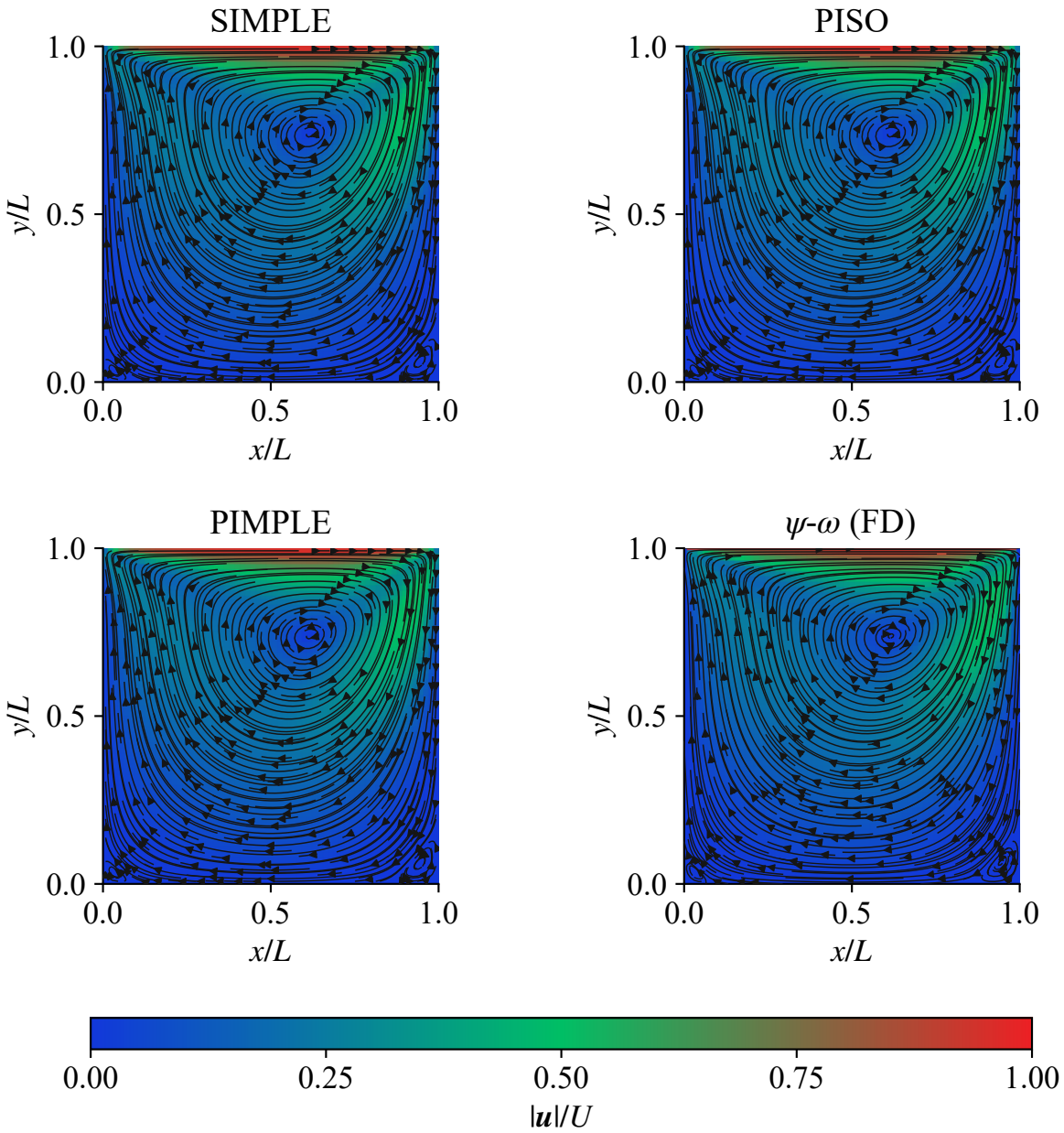


Figure 13: Retained Re=100 speed and velocity streamlines at h=1/64. All panels use the same physical color limits. Streamline arrows show circulation direction; their density does not measure vorticity.

Speed is $|\boldsymbol{u}| = \sqrt{u^2 + v^2}$ and signed vorticity is $\omega = \partial v/\partial x - \partial u/\partial y$. The moving lid drives a clockwise primary circulation and a lower-cavity return flow. Black streamlines follow computed velocity; they are not vorticity contours. Small corner eddies depend on resolution and convergence as well as display density.

17 continued Signed vorticity and circulation

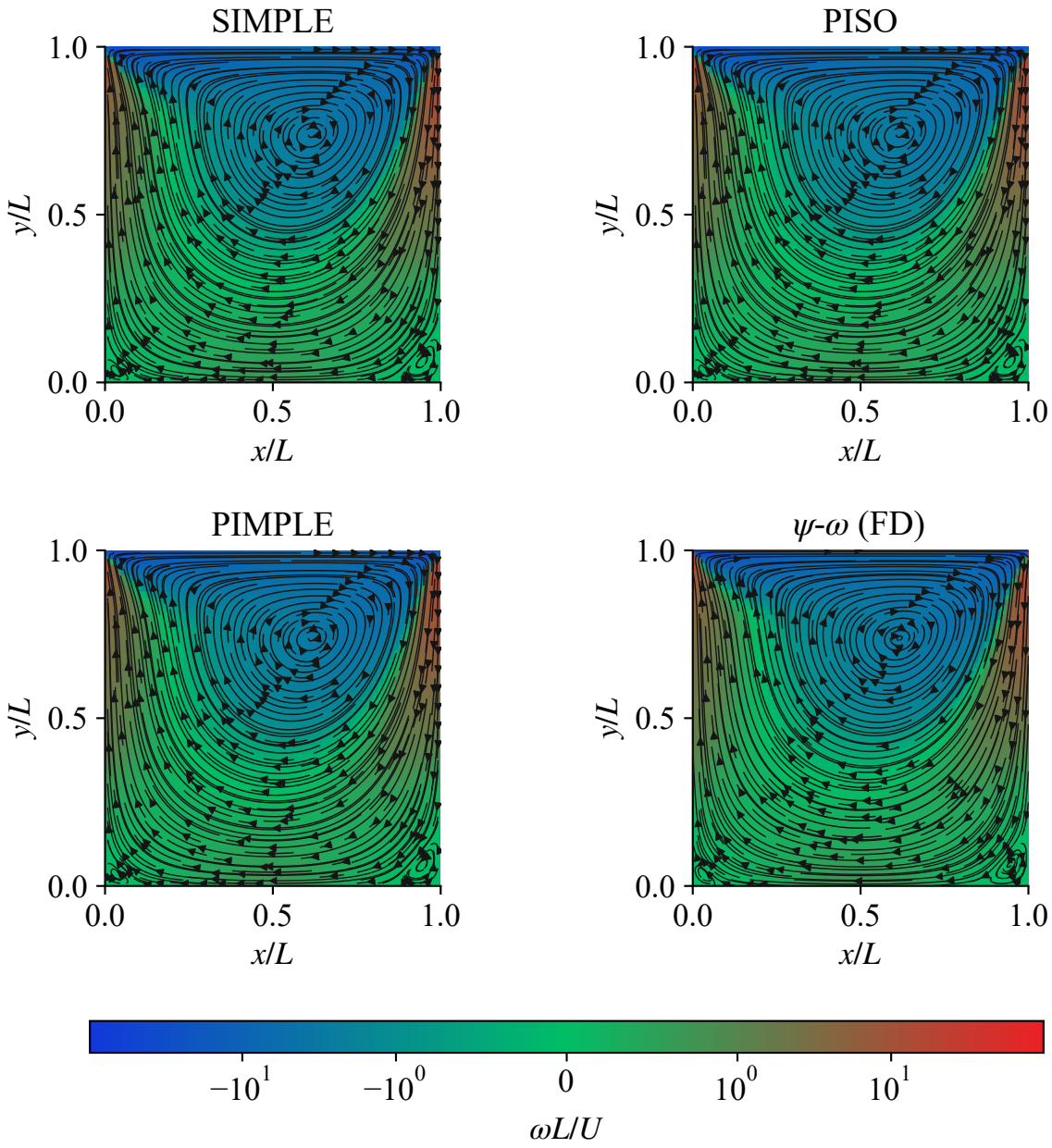


Figure 14: Retained Re=100 signed vorticity with velocity streamlines at h=1/64. One symmetric logarithmic color scale is shared by all panels, linear for absolute vorticity up to 1. The displayed colorbar values are nondimensional physical vorticity, including wall peaks.

Signed fields use a common symmetric-logarithmic color scale across methods, resolving interior values while retaining strong wall gradients. Colorbar values remain physical nondimensional field values. Similar converged fields are expected when the same discrete steady equations are solved; the comparison is useful for checking consistency.

18 Pressure and native convergence histories

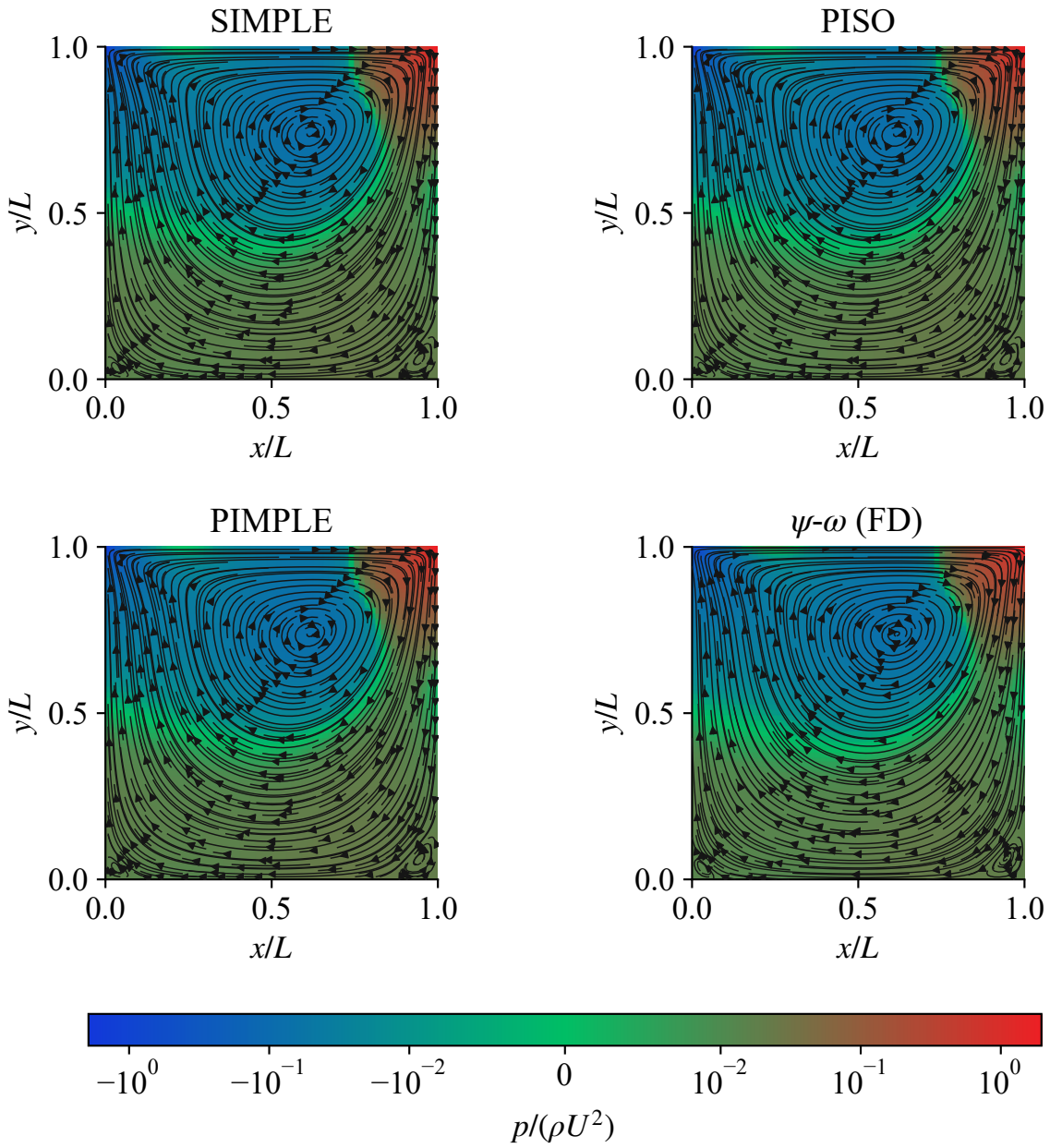


Figure 15: Retained Re=100 zero-mean pressure and velocity streamlines at h=1/64. One symmetric logarithmic color scale is shared, linear for absolute pressure up to 0.01. FD pressure is reconstructed after the vorticity solve. No independent pressure benchmark is claimed.

All pressures are reported with zero mean. Pressure gradients, rather than the arbitrary mean, affect velocity. Differences near the discontinuous lid corners should not be interpreted as reference-pressure validation. Ghia supplies velocity data, not pressure data.

18 continued Native convergence histories

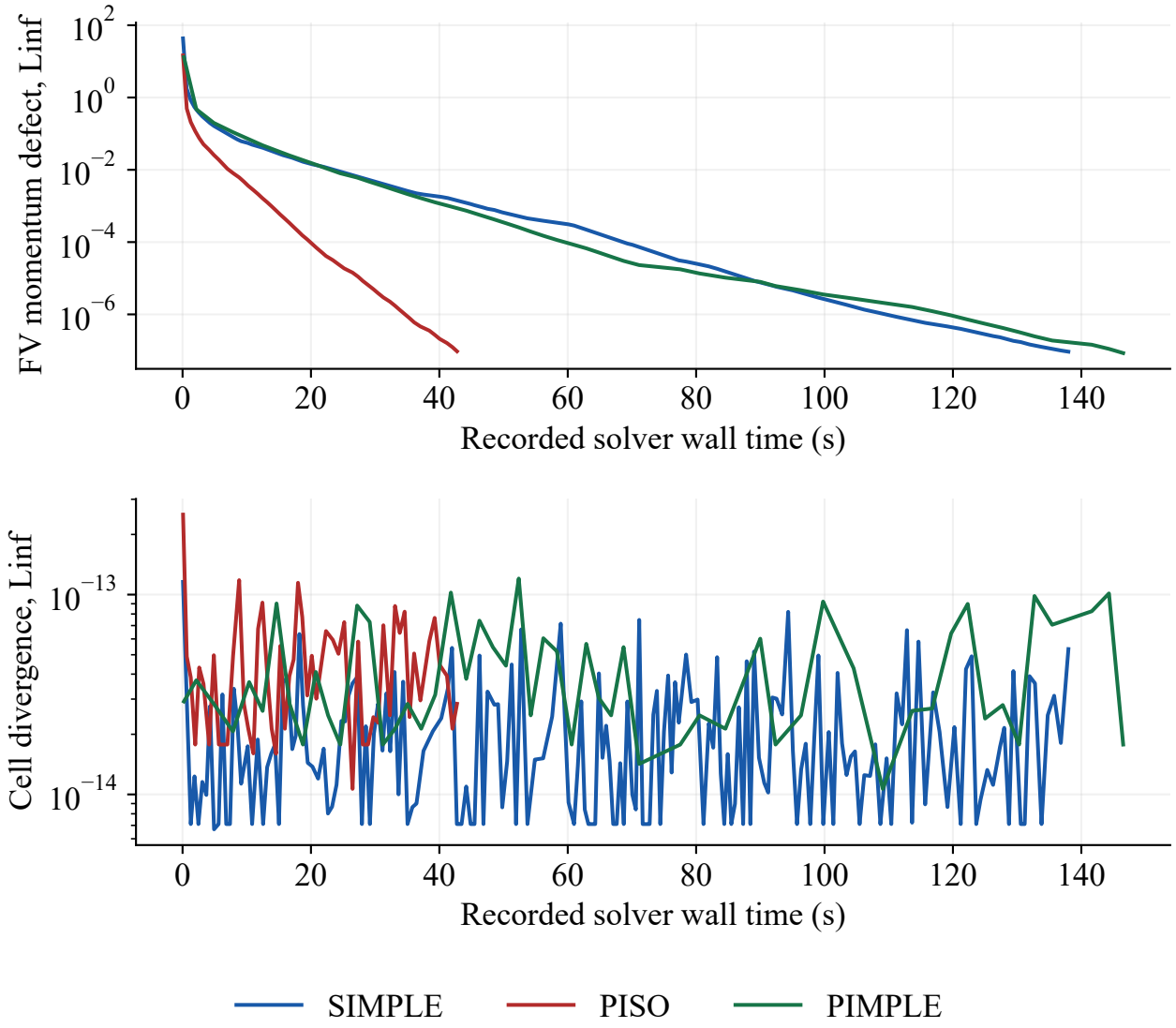


Figure 16: Retained Re=100 finite-volume convergence histories at h=1/64. The horizontal coordinate is recorded solver-call wall time from one local run per method. Momentum and continuity are separate defects. The streamfunction-vorticity solver uses a different native stopping norm and is included in the adjacent timing tables, not on these FV residual axes.

The finite-volume histories evaluate momentum and mass balance. The streamfunction solver's native vorticity-update norm is kept separate in the notebook because it is a different quantity. Timing comparisons still include that formulation. A small relaxed update alone is not a momentum-equation convergence test.

19 Accuracy and recorded cost at $Re = 400$ and 1000

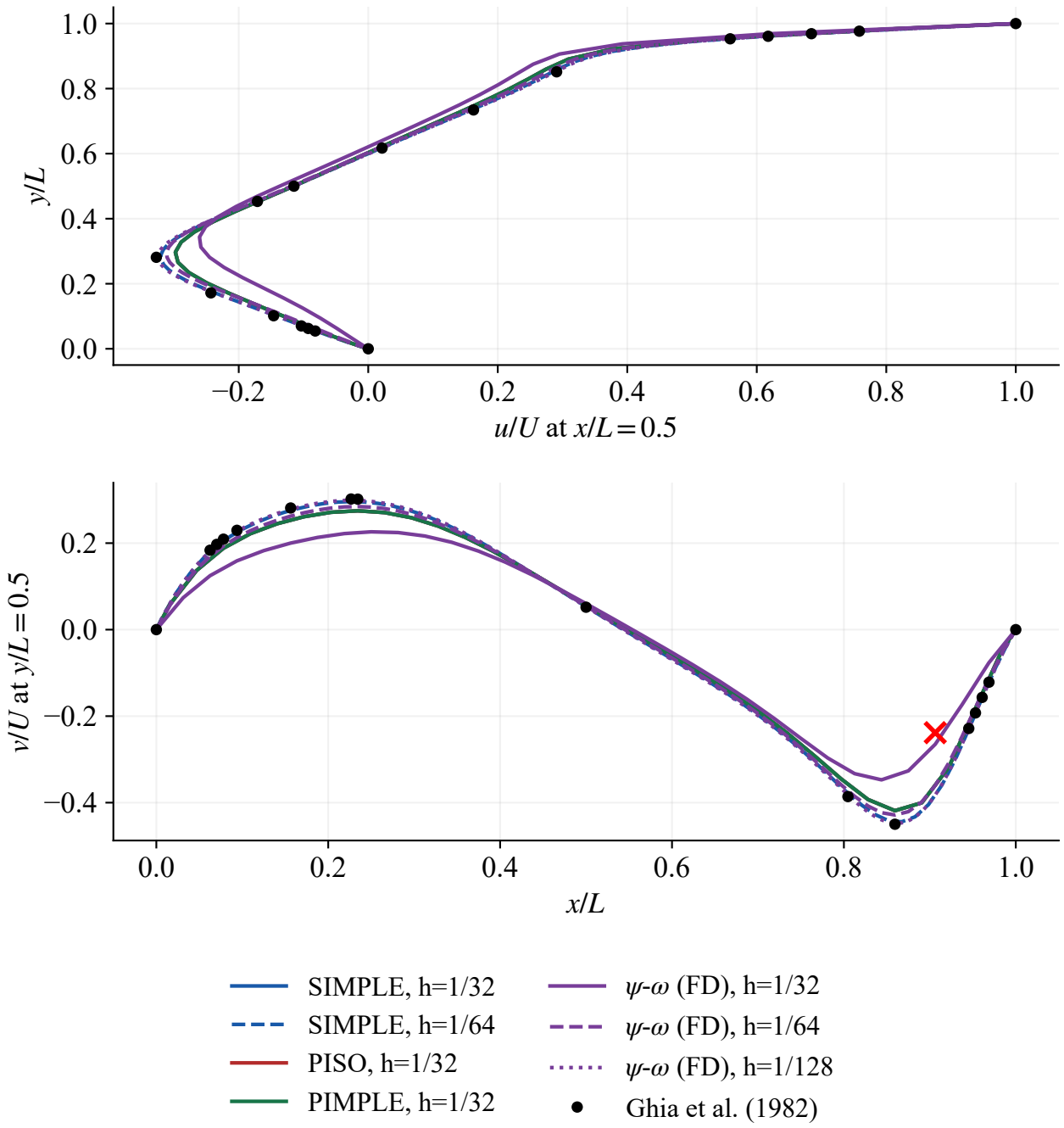


Figure 17: Retained Re=400 centerline velocities. Color identifies formulation and line style identifies spacing. Black markers are Ghia samples; the red cross retains the isolated published horizontal-profile anomaly at x/L=0.9063, excluded only from the qualified error norm. Selected fine cases do not constitute a complete refinement matrix.

The retained study includes all four methods at $h = 1/32$, plus selected finer meshes. A converged coarse-grid streamfunction result can have large reference error. At $Re = 1000$, its 128-cell-spacing relative profile errors are approximately 2.0% for u and 1.7% for v, using $\|f_{\text{computed}} - f_{\text{Ghia}}\|_2 / \|f_{\text{Ghia}}\|_2$ at the tabulated samples.

19 continued Centerline accuracy at $Re = 1000$

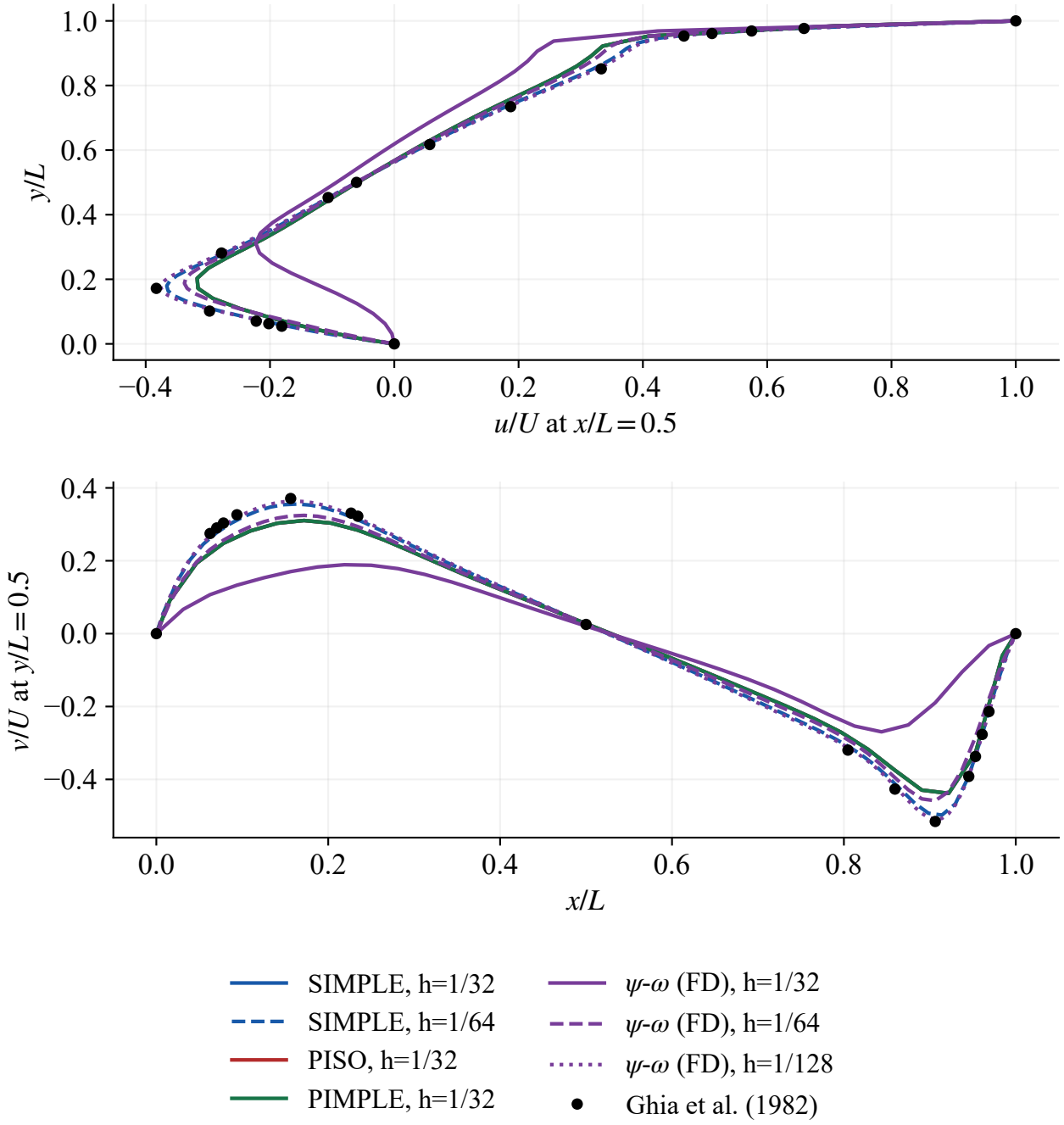


Figure 18: Retained Re=1000 centerline velocities. Color identifies formulation and line style identifies spacing; black markers are Ghia et al. (1982) samples. Finer FD solutions show the effect of spatial resolution separately from the coupling algorithm.

19 continued Recorded higher-Reynolds-number cost

Formulation	*Re*	Cells per side	Recorded seconds
PIMPLE	400	32	96.949
PISO	400	32	22.278
SIMPLE	400	32	11.793
Streamfunction–vorticity	400	32	78.475
SIMPLE	400	64	71.061
Streamfunction–vorticity	400	64	28.382
Streamfunction–vorticity	400	128	70.928
PIMPLE	1000	32	175.140
PISO	1000	32	55.302
SIMPLE	1000	32	12.459
Streamfunction–vorticity	1000	32	175.065
SIMPLE	1000	64	62.141
Streamfunction–vorticity	1000	64	69.579
Streamfunction–vorticity	1000	128	171.387

For an error vector z over the reference samples, $\|z\|_2 = \sqrt{\sum_j z_j^2}$; j labels a sampled location. These are recorded sequential local solver-call timings. Reference errors, mesh spacing and native stopping definitions accompany them. The $Re = 400$ all-point v error retains the isolated published entry; a separately labeled sensitivity metric omits that sample without replacing its value.

20 Mesh refinement and recirculation

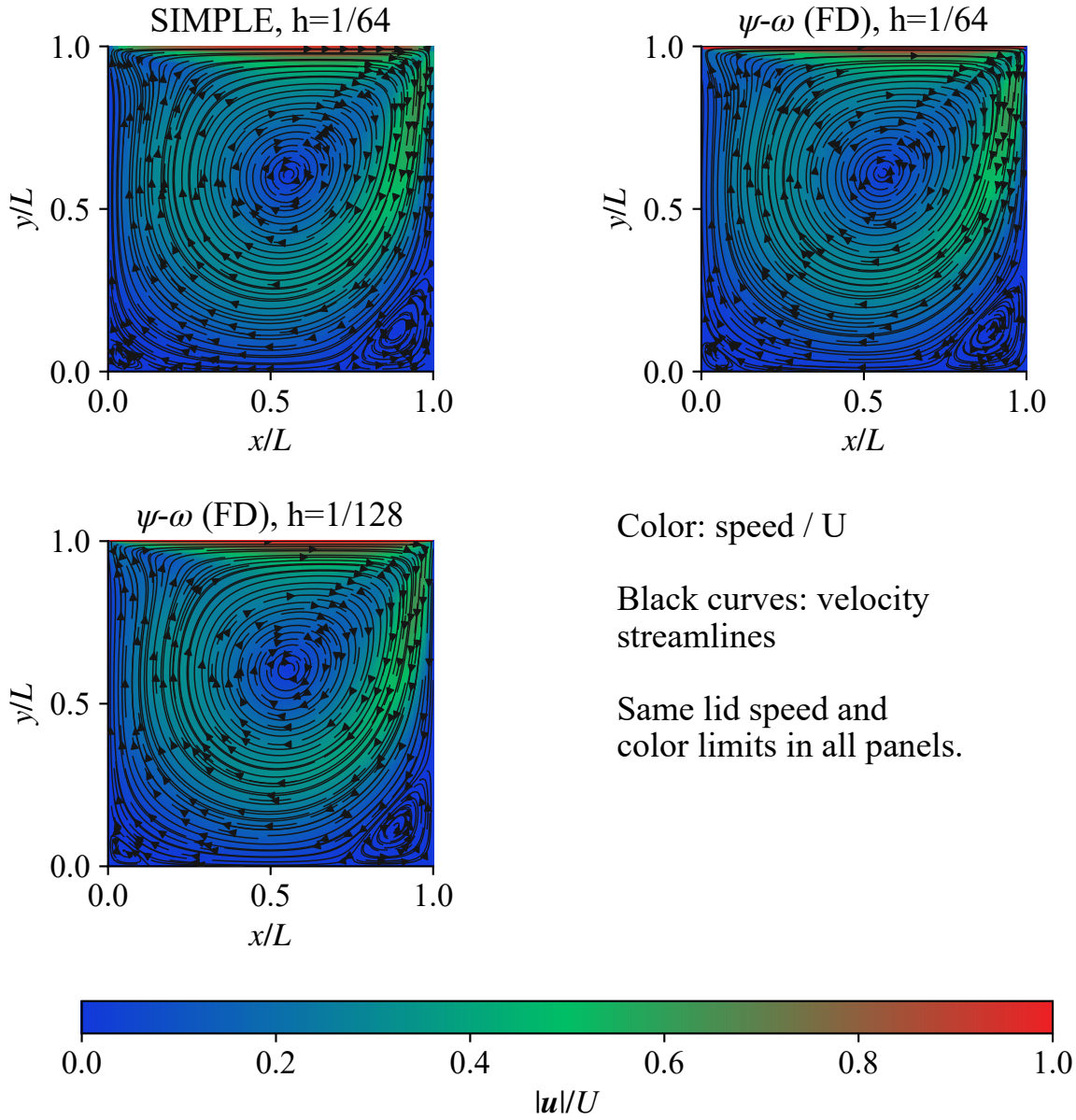


Figure 19: Retained Re=400 speed and computed velocity streamlines: SIMPLE at h=1/64, and independent streamfunction-vorticity solutions at h=1/64 and 1/128. The color scale and lid speed are common. Small recirculation structures must be supported by the resolved velocity field.

Each panel states Reynolds number, formulation and spacing. The sampled primary-vortex location and resolved corner circulation change with refinement. More inner pressure corrections cannot reproduce an unresolved boundary layer or secondary vortex.

The fine-grid study is selected rather than a complete four-method refinement matrix. A result at one mesh does not demonstrate asymptotic order. Native steady-equation checks, Ghia errors and recorded initial failed attempts remain available with the data.

20 continued Refined circulation at $Re = 1000$

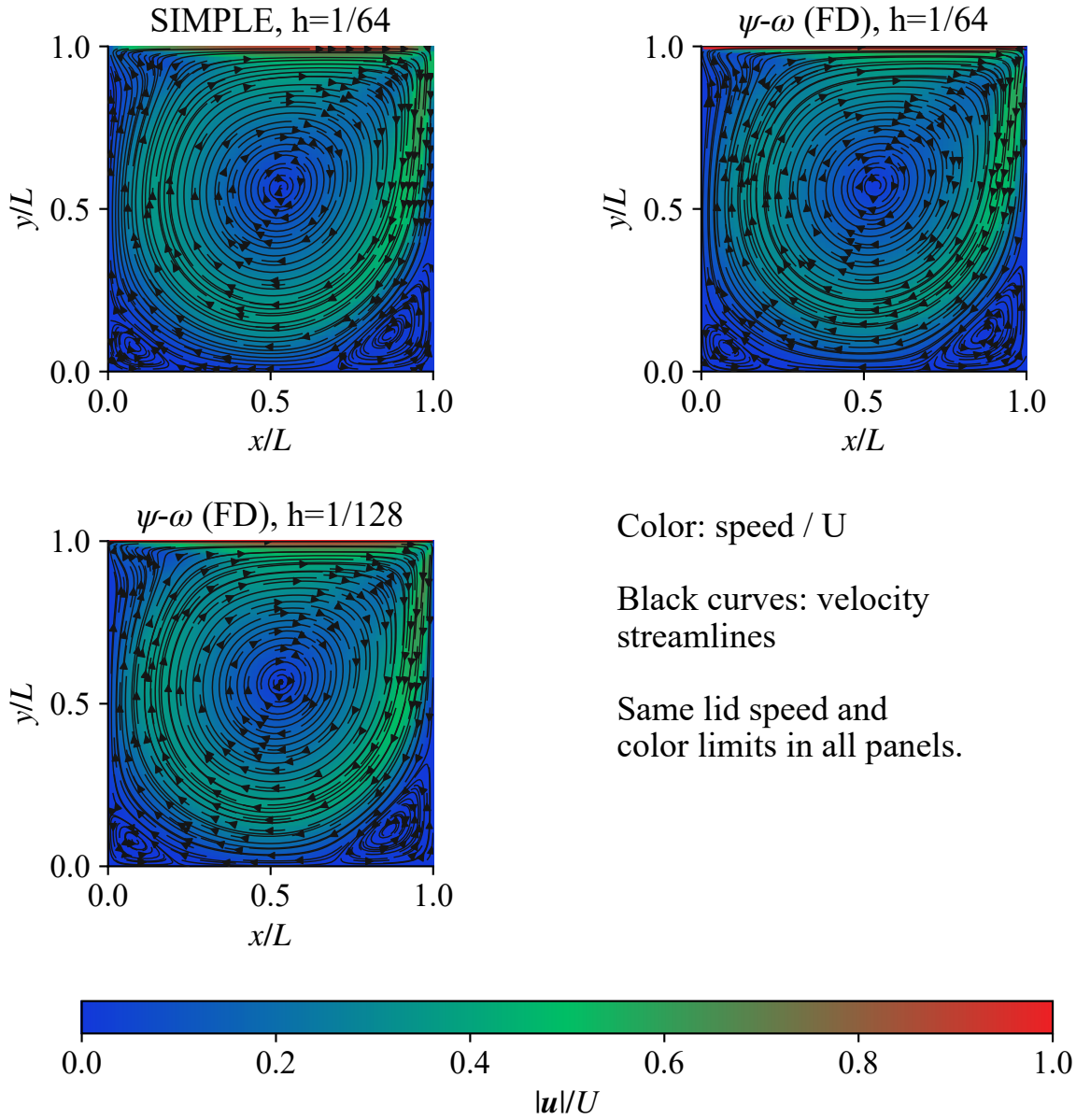


Figure 20: Retained Re=1000 speed and computed velocity streamlines for the same selected refinement cases. The finer grid changes resolved boundary layers and corner circulation. Additional pressure corrections alone cannot recover spatial structures absent from the mesh.

Computational cost. Fine grids and higher Reynolds numbers may require substantially longer runtimes and more memory. Students should begin with the supplied coarse case and estimate cost before selecting 64 or 128 cells on a laptop or Colab runtime.

21 A measured nonlinear coupling effect in the cavity

Method	Δt	Outer	Inner	Nonlinear momentum defect	Continuity defect
PISO	0.01	1	1	1.61×10^{0}	5.90×10^{-14}
PISO	0.01	1	2	1.78×10^{-1}	4.20×10^{-14}
PISO	0.01	1	4	1.78×10^{-1}	4.26×10^{-14}
PIMPLE	0.01	3	2	7.26×10^{-6}	5.23×10^{-14}
PIMPLE	0.01	6	2	1.14×10^{-12}	6.84×10^{-14}
PISO	0.05	1	1	5.13×10^{0}	7.27×10^{-14}
PISO	0.05	1	2	1.11×10^{0}	2.38×10^{-14}
PISO	0.05	1	4	1.12×10^{0}	2.96×10^{-14}
PIMPLE	0.05	3	2	3.43×10^{-3}	4.92×10^{-14}
PIMPLE	0.05	6	2	3.75×10^{-7}	4.93×10^{-14}
PISO	0.2	1	1	9.34×10^{0}	1.37×10^{-13}
PISO	0.2	1	2	2.39×10^{0}	8.88×10^{-15}
PISO	0.2	1	4	2.47×10^{0}	1.14×10^{-14}
PIMPLE	0.2	3	2	4.92×10^{-2}	9.77×10^{-15}
PIMPLE	0.2	6	2	1.46×10^{-4}	3.38×10^{-14}

Fifteen configurations start from rest at $Re = 100$ on 32 cells per side. Each takes one backward-Euler step. At each fixed Δt, the study varies inner pressure corrections and outer coefficient updates.

The nonlinear step diagnostic rebuilds convection from the final velocity, while the frozen-predictor diagnostic uses the coefficients assembled for the predictor. Extra PISO corrections can reduce pressure–velocity inconsistency without eliminating the error from frozen nonlinear coefficients. PIMPLE also updates those coefficients, with additional momentum assembly and solves.

All retained cases in this one-step study satisfy a cell-divergence defect below 10^{-9}. The remaining nonlinear momentum defects distinguish the correction strategies. Their differences demonstrate algebraic coupling behavior for this setup, not the accuracy of a transient trajectory.

Each panel has its own logarithmic vertical range. For a temporal-accuracy study, reduce Δt and compare at matched physical times. More correctors and a stable larger time step do not establish an accurate startup history or a universal wall-time advantage.

22 Notebook controls and student analysis

The notebook presents the same derivations and runnable algebraic examples, followed by retained CFD results. Fresh CFD runs are opt-in so the default lesson can be inspected without repeating every expensive calculation.

Control	Purpose
`RUN_NEW`	Fresh baseline calculation at $Re = 100$
`RUN_HIGHER_RE`	Fresh experimental higher-Re calculation
`HIGHER_METHOD`	One method or all four formulations
`HIGHER_CELLS`	Number of cells per side / matched FD spacing
`RUN_STREAMFUNCTION`	Independent FD refinement experiment

Conservation. Derive the four-face cell pressure equation and verify the two-cell example. Explain why a zero global flux does not guarantee zero cell imbalance. Add a constant pressure offset and check that velocities do not change.

PISO. Reproduce both corrections of the frozen-matrix example. Identify which quantities change between correctors. Explain why zero divergence after the first correction does not imply zero momentum defect.

PIMPLE. Reproduce the three outer passes of the nonlinear example. Evaluate the nonlinear defect using coefficients rebuilt from the final velocity. Explain why the old-time source must stay fixed. Compare one outer pass with PISO.

Cavity comparison. Start at $Re = 100$ and matching mesh spacing. Record final cell divergence, unrelaxed momentum defect, both Ghia profile errors, runtime, and solve counts. Change one coupling setting at a time; keep spatial equations and final accuracy targets fixed.

Higher Reynolds number. At $Re = 400$ and 1000, separate spatial refinement from coupling changes. Retain failed configurations. State the cost before proposing a fine-grid run. Native FD and FV residual magnitudes must not be interpreted as the same error.

Transient extension. Specify a final physical time, run a sequence of halved time steps, and compare common field or phase errors. Keep the time scheme fixed when comparing coupling algorithms. The current steady SIMPLE iteration is not a transient trajectory.

23 References

The textbook treatments develop finite-volume conservation, staggered pressure–velocity coupling and pressure correction. The paper and official algorithm documentation provide the PISO origin and practical inner/outer-loop distinctions.

[1] S. V. Patankar (1980). *Numerical Heat Transfer and Fluid Flow*. Hemisphere/Taylor & Francis. Publisher. Foundational pressure correction, staggered grids and SIMPLE.

[2] H. K. Versteeg and W. Malalasekera (2007). *An Introduction to Computational Fluid Dynamics: The Finite Volume Method*, 2nd ed. Pearson. Publisher. Finite-volume transport and pressure–velocity coupling.

[3] J. H. Ferziger, M. Perić and R. L. Street (2020). *Computational Methods for Fluid Dynamics*, 4th ed. Springer. doi:10.1007/978-3-319-99693-6. Incompressible-flow solution methods and numerical errors.

[4] F. Moukalled, L. Mangani and M. Darwish (2016). *The Finite Volume Method in Computational Fluid Dynamics*. Chapter 15, "Fluid Flow Computation: Incompressible Flows," pp. 561–654. Chapter DOI. Staggered/collocated arrangements and SIMPLE/PISO derivations.

[5] R. I. Issa (1986). "Solution of the implicitly discretised fluid flow equations by operator-splitting." *Journal of Computational Physics* 62(1), 40–65. doi:10.1016/0021-9991(86)90099-9. Original PISO formulation.

[6] CFD Direct, *OpenFOAM User Guide v13*, sections 4.6.7–4.6.8. Solution and algorithm control. Pressure reference, correctors and outer correctors. The teaching solver's fixed-loop MAC implementation is specified in the preceding equations; software variants need not have identical defaults.

[7] OpenCFD, *Pressure–velocity coupling*. Official equation decomposition. Momentum diagonal and neighbor-source representation.

[8] U. Ghia, K. N. Ghia and C. T. Shin (1982). "High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method." *JCP* 48, 387–411. doi:10.1016/0021-9991(82)90058-4. Velocity benchmark samples.

The examples use a staggered grid. Collocated-grid interpolation and Rhie–Chow stabilization are separate topics and are not implemented by this solver.

23 continued Guided reading and video lectures

1. From momentum to a pressure equation. Chris Greenshields and Henry Weller, *Notes on Computational Fluid Dynamics: General Principles*, section 5.10, Pressure–velocity coupling. Read alongside Sections 4–7 of this lecture. Follow the momentum diagonal, the face-flux correction and substitution into continuity. Their software notation differs from the staggered scalar equations used here.

2. SIMPLE and under-relaxation. The same authors' section 5.12, Steady-state solution, is a companion to Section 8. Identify the momentum predictor, conservative flux correction and next nonlinear iteration. Compare it with Aidan Wimshurst's Fluid Mechanics 101 video, *[CFD] The SIMPLE Algorithm (to solve incompressible Navier–Stokes)*. Use the diagram in this lecture to track where each unknown is stored while following the pressure-correction sequence in the video.

3. PISO within a physical time step. Read section 5.19, Transient solution, alongside Sections 9–11. Then watch Fluid Mechanics 101, *[CFD] The PISO Algorithm*. Focus on what is updated before the next pressure solve, and what remains fixed during the time step. Non-orthogonal corrections discussed in the video are a separate mesh issue; the present uniform Cartesian grid does not require them.

4. PIMPLE and nonlinear outer iterations. Read section 5.21, The PIMPLE algorithm, alongside Sections 12–14. Compare the outer coefficient updates with the inner corrections. Distinguish time-accurate integration from a pseudo-transient route to a steady state; more correctors do not replace a time-step refinement study.

5. Connect the loops to practical controls. The official OpenFOAM v13 User Guide, Solution and algorithm control, explains pressure references, `nCorrectors` and `nOuterCorrectors`. Map these names to the loops before reading software configuration examples. This Python lesson uses its own MAC-grid implementation, so software-specific interpolation details and defaults should not be transferred without checking.

The videos are linked as companion explanations. The author's lecture catalogue lists the SIMPLE and PISO lectures and distinguishes the YouTube videos from separately sold slide collections. No paid slides are required for this lesson.